



Federal Office
for Information Security

5G Risk Analysis

Detailed Analysis: **Management System**



Version History

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1 List of Abbreviations

3GPP	3rd Generation Partnership Project
AKA	Authentication Key Agreement
AS	Access Stratum
AMF	Access Mobility Function
API	Application Programming Interface
BBU	Base Band Unit
BSS	Business Support System
CN	Core Network
CM	Configuration Management
COTS	Commercial-Of-The-Shelf
CU	Centralized Unit
DoS	Denial of Service
DU	Distributed Unit
ETSI	European Telecommunications Standard Institute
FCAPS	Fault Management (F), Configuration (C), Accounting (A), Performance (P) and Security (S)
FM	Fault Management
gNB	gNodeB
HLR	Home Location Register
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
HW	Hardware
IMEI	International Mobile Equipment Identity
IMS	IP Multimedia System
IMSI	International Mobile Subscriber Identity
ISO	Organization for Standardization
IRP	Integration Service Reference PoinP
LI	Lawful Interception
MANO	Management and Orchestration
MEC	Mobil Edge Computing
MGMT	5G Management Systeme
MNO	Mobile Network Operator
MnS	Management Services
NAS	Non-Access Stratum
NFVI	Network Function Virtualization Interface

NG	Next Generation
NRM	5G Network Resource Model
MDA	Management Data Analytics
MDAS	Management Data Analytics Services
OMA	Orchestration, Management, and Automation
OSS	Operations Support System
OWASP	Open Web Application Security Project
PCI	Public Cell Identifier
PM	Performance Management
RA	Risk Analysis
RAN	Radio Access Network
RCS	Rich Communication Services
SBA	Service Base Architecture
SBMA	Service Based Management Architecture
SIM	Subscriber Identity Module
SMF	Session Management Function
SON	Self Organizing Network
TAC	Tracking Area Code
TLS	Transport Layer Security
UDM	Unified Data Management
UE	User Equipment
VoLT	Voice over LTE
VoNR	Voice over New Radio
VoWiFi	Voice over WiFi

2 Introduction and Purpose

2.1 Intention

The intention of the document is to dive deep into the analysis of risk scenarios where the **management layer** is the attacked or failing subsystem of a 5G network.

This document shows the current view on the probability that any risk scenarios could become reality.

The study is only available in English language, as the target group is primarily the international IT security community. The study is also in the context of English publications on this topic.

2.2 Link to Framework Document

This document is linked to the framework document and provides relevant information about the **management system** in 5G mobile networks and a probability assessment of relevant risk scenarios described in the framework document. The overall risk assessment for each risk scenario consists of the results of this analysis and the impact described in the framework document. The overall risk assessment for the management system is summarized in Chapter 6 of this document and in the framework document.

3 Methodology

The methodology used for the 5G Risk Analysis (5G-RA) is based on PASTA. In this document and the related documents, we will refer to the methodology as 5G-RA.

This chapter explains the seven steps of the 5G-RA methodology and assigns them to the corresponding PASTA steps.

Step 1: Definition of Risk Scenarios

In the first step of the 5G-TARA, the 5G network needs to be analysed from the following three perspectives:

1. Public Mobile Network Operator (MNO) Perspective,
2. Private 5G Network Operator Perspective,
3. User Perspective.

Several risks with a significant impact on the society or public safety have been identified by the legislative in the past. To mitigate these risks, laws were set into force, e.g. the legislative to ensure emergency calls are available (TKG §165). Most of these laws define fines for mobile network operators if the regulative obligation is not followed.

These perspectives are explained in detail in Chapter 6 of the framework document. The goal is to identify the possible risks that the 5G network might be exposed to from each perspective.

Step 2: 5G System Diagram

In step 2, a detailed system diagram of the parts of a 5G network needs to be created. This diagram will serve as a basis for the following steps.

Step 3: Refinement of Detailed 5G Subsystem Diagrams

In this step, more technical definitions and explanations of the relevant 5G subsystems are introduced. For example, various network functions are part of the 5G core network. This level of detail serves as a basis for the following steps.

Step 4: Collection of Known 5G Threats

In this step, publicly known 5G specific threats are collected. For this purpose, relevant studies and input documents are collected. This is the basis for the identification of threats in the next step.

Step 5: Search of 5G Vulnerabilities

Step 5 includes the analysis and identification of relevant threats for each risk scenario. The findings of this step are part of the corresponding Detailed Risk Analyses.

Step 6: Examination of Identified Threats

In this step, all threats that were found in step 5 are examined on whether there is a possibility for this threat to be realized. Hypothetical scenarios are used as well as references to successful attacks, which were performed in the past.

Step 7: Risk Calculation

The last step of the methodology is a calculation of the risk for all threats that were identified in step 5 and are still considered relevant after step 6 was carried out. The overall results are presented in Chapter 6 of this document.

The following table assigns the steps of the 5G-RA to the steps of PASTA and gives an overview of the document containing the results of each step.

Table 1: Assignment of 5G-RA steps to PASTA-steps

#	5G-RA methodology	Document Layer	PASTA
1	Definition of Risk Scenarios	Framework	Define Objectives
2	5G System Diagram	Framework	Define Technical Scope
3	Refinement of Detailed 5G Subsystem Diagrams	Framework & Detailed Risk Analysis	Application Decomposition
4	Collection of Known 5G Threats	Detailed Risk Analysis	Threat Analysis
5	Search of 5G Vulnerabilities	Detailed Risk Analysis	Vulnerability and Weakness Analysis
6	Examination of the Identified Threats	Detailed Risk Analysis	Attack Modelling
7	Risk Calculation	Framework & Detailed Risk Analysis	Risk and Impact Analysis

4 5G Network Model: Management System

The 5G network model introduced in the subsequent sections is used to split the 5G-RA into multiple Detailed Risk Analyses. The model adheres to the relevant standards (e.g 3GPP standards) as much as possible. The 5G network model comprises a variety of interconnected subsystems that can be categorized from a high-level perspective, as shown in Figure 1. In the context of this risk analysis, it is assumed that the network under consideration is a 5G Standalone system. In this case, the Radio Access Network (RAN) as well as the Core are 5G-specific subsystems.

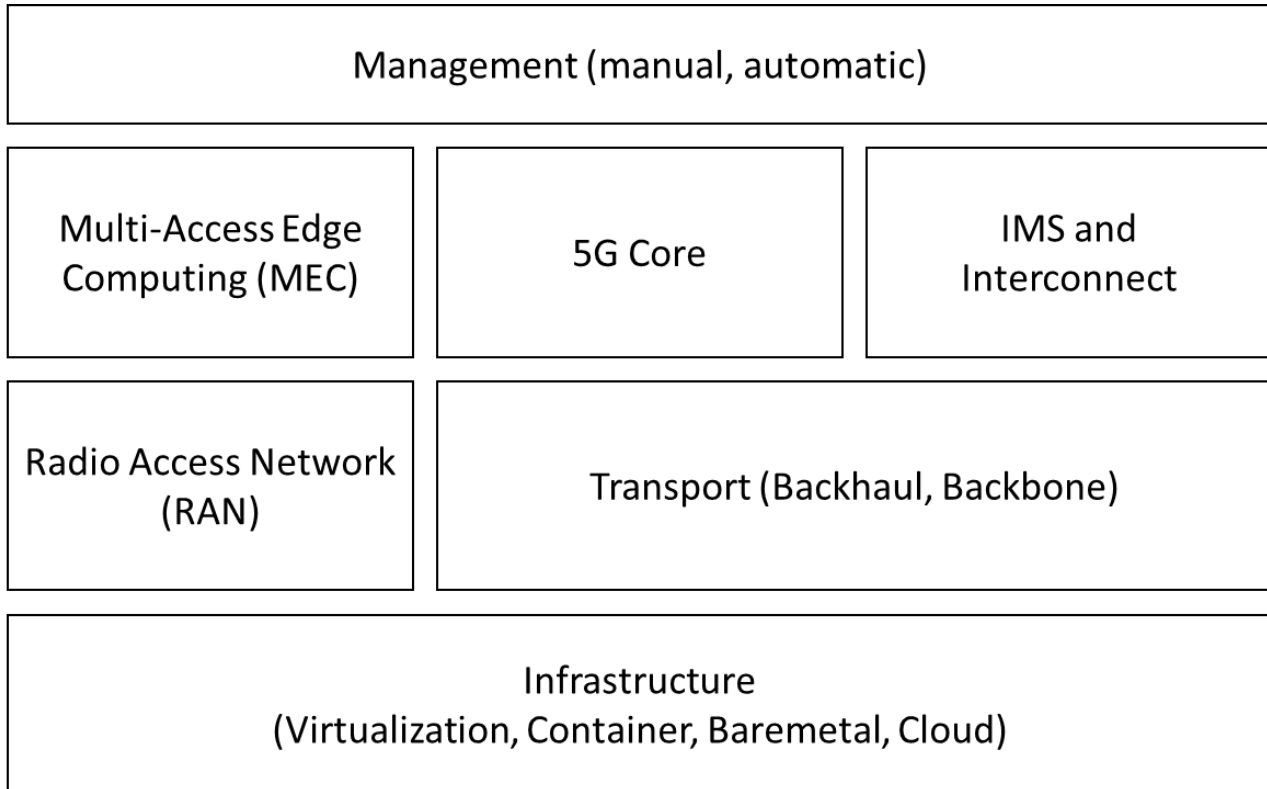


Figure 1: 5G Network Model: High level view

Other subsystems are not specifically designed for the 5G architecture and used either to operate the 5G system or to connect it to other internal and external networks.

4.1 Management in 5G

For 5G management, the 3GPP defines a Service Based Management Architecture (SBMA) in TS 28.533 Release 16. The basic building blocks for the SBMA are Management Services (MnS). An entity that provides a service is called an MnS producer. An entity that consumes a service is called a MnS consumer. Communication takes place via a standardized service interface. 3GPP TS 28.319 specifies a RESTful HTTP(S) approach for establishing connections and exchanging data. TLS is used for mutual authentication between the management services and management consumer. Role based access management is standardized as using NETCONF/YANG. The MnS are to replace the Integration Service Reference Point (IRP) information services used for data exchange. The SBMA with the MnS is currently in the specification phase and is not yet reflected in all management specifications for the respective domains. The specifications currently focuses on the areas of Fault Management (FM), Performance Management (PM) and Configuration Management (CM).

Based on current information, the specification of the SBMA, which was introduced with the 5G standard for management tasks, is still under development. While the Service Base Architecture (SBA) provides much more detailed and binding requirements, many aspects of the SBMA standards seem to be chiefly of an informative nature. It is currently uncertain to what extent the specifications will delve into the technical implementation details. Based on the history of the 3GPP technical specifications, it can be expected that there will be considerable freedom for company-specific implementations. Some of the documents for the individual domains do not yet mention 5G and often still refer to 2G, 3G or LTE.

The 3GPP study 3GPP TR 28.925 provides an overview of the potential enhancement of the 5G service-based management architecture compared to pre-5G variants, but being a study it is not normative in nature.

4.1.1 Management Domains

The 5G network consists of several domains, that must each be managed separately. However, there are also dependencies between the domains that require handling by a higher-level management system (OSS). Similarly, not all domains need to be present in a given 5G network. The MEC domain can be considered optional and the IMS domain may not be necessary, for example in a non-public network (private network) that is mainly used for production automation.

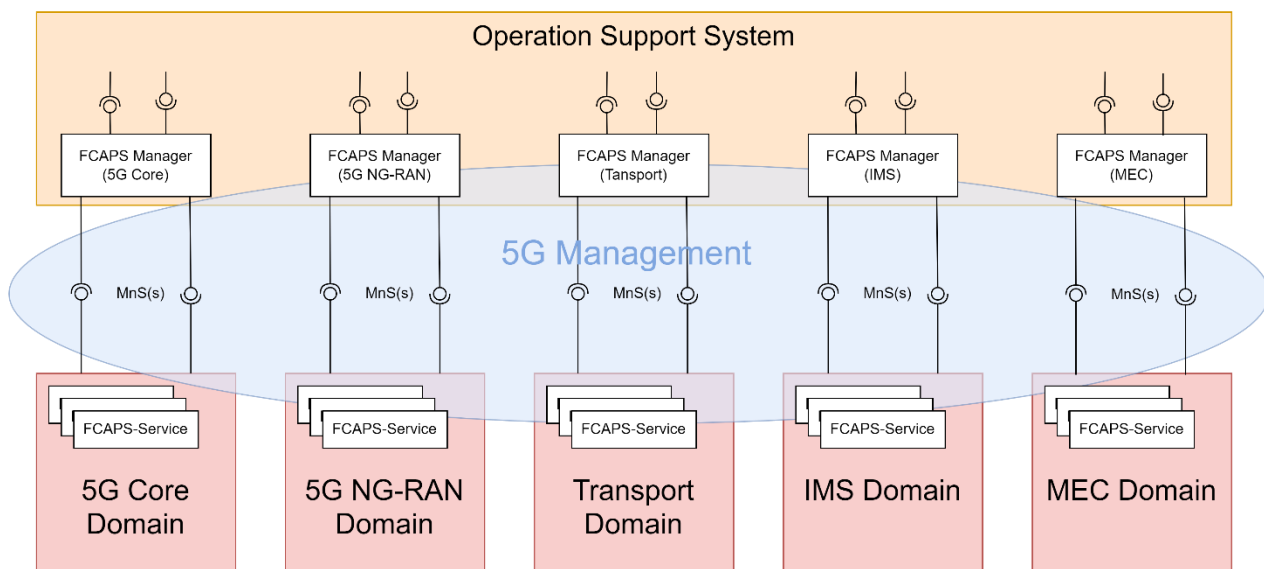


Figure 2: 5G Management Domains

4.1.1.1 5G NG-RAN

The task of the radio access network (RAN) is to enable radio communication between user equipment (UE) and the base station (BBU). It also controls the communication channels that are established via the antennas on the air interface and realizes uninterrupted mobility control for the end devices. The central component of the RAN is the Next Generation NodeB (gNB). The gNB can be subdivided into a Central Unit (CU) and a Distributed Unit (DU). The CU is a logical node that includes functions like transfer of user data, mobility control, radio access network sharing, positioning, session management etc. The DU includes a subset of the gNB functions.

The RAN is mentioned in TS 28.533 as an example of management via the SBMA. The 3GPP specification TS 28.622, which is relevant for RAN management, currently still describes management based on IRP Information Services, but already indicates that the document applies to deployment scenarios that use the Service Based Management Architecture (SBMA) defined in TS 28.533.

4.1.1.2 5G Transport

The transport network provides the connection from the RAN to the core network (CN), as well as the connection between the gNB, CU and DU, depending on the implementation. The transport network is most often implemented using fiber optic lines, but wireless communication can also be used. The transport network must provide features such as network slicing to enable, for example, ultra high bandwidth for mMTC scenarios and ultra-low latency for URLLC scenarios.

All currently available 3GPP standards describing the management for the transport domain refer to the use of IRP Information Services. At present, there seems to be no specification that describes management using SBMA. However, since according to 3GPP TS 28530 the IRP service is intended to be converted to the use of MnS, it is to be expected that management will be using SBMA in the future.

4.1.1.3 5G Core

The 5G Core (5GC) is a fundamental building block of the 5G network. It is functionally subdivided into control plane and data plane sub-areas. The control plane provides functions at the control level, subscriber data management (UDM) and policy management (PCF), authentication and authorization (AUSF), signaling management (Session Management Function, SMF), and registration and mobility management (Access and Mobility Management Function, AMF). The User Plane provides services for voice and data transmission. The functional entities of the 5G network, referred to as Network Functions, are fully realized in software and virtualized.

The 3GPP specifications for the management of the IMS currently all refer to the use of IRP services. In 3GPP TS 28530, the IMS domain is listed as an example of 5G management based on the SBMA. The 3GPP TS 23.228 for management in the IMS domain does not yet describe any concepts that would allow any reliable conclusions to be drawn about integration of management functions into the SBMA of the 5G standard.

4.1.1.4 5G IMS and interconnect

The IP Multimedia Subsystem (IMS) is a telecommunications service framework that delivers multimedia services like voice services (telephony calls, emergency calls, etc.), video chat, video conferencing, and Instant Messaging-based multimedia services (VoNR, VoLTE, VoWiFi, RCS, etc.) over IP Networks. The first specification for the IMS dates back to 2003 and was created by the 3rd Generation Partnership Project Panel (3GPP). The original area of application for the IP Multimedia Subsystem was the service network of the 3G UMTS mobile network. IMS is now also used in 4G LTE and 5G networks. The 3GPP TS 26.114 specification provides a comprehensive overview of the functionality of the IMS.

All currently available 3GPP standards describing the management for the core network domain refer to the use of IRP Information Services. At present, there is apparently no specification that describes management using SBMA in detail. The CN is listed in the 3GPP as an example for management using SBMA.

4.1.1.5 MEC

Multi-access Edge Computing (MEC) offers a framework for running services at the edge of a network, providing ultra-low latency and high bandwidth. The MEC provides a platform to host user applications, specifying a set of standardized interfaces through which applications can access information from a network (e.g. User Location, Quality of Service Information, Routing) without needing to be aware of the underlying network technology, whether it be mobile networks, WLAN, or fixed access networks. The ETSI

MEC specifications also describe the management and orchestration of the MEC platform and applications, on a host level as well as on a system wide level.

Efforts have been made to harmonize the ETSI MEC and the 3GPP Edge Computing specifications. Both the 3GPP TS 28.538 specification for edge computing management and the ETSI standards for the MEC describe a management procedure that corresponds to that of the SMBA.

4.1.2 3GPP 5G Management

The Standalone 5G core network is organized as a service-based architecture. It is made up of various function modules, known as network functions. These are virtualized via the "Network Functions Virtualization Framework" standardized by the ETSI. The following image shows the structure of this framework, with the Management and Orchestration part (MANO) on the right-hand side and the Network Function Virtualization Infrastructure (NFVI) in the bottom of the image. MANO and NFVI are not within the scope of this study.

The network functions are housed either in their own virtual machines or software containers. As a result, they work in isolation from each other and can be dynamically allocated with the appropriate resources for their respective tasks. To ensure correct functioning and cooperation, network functions must be configured individually depending on their task, and their current status must be continuously monitored.

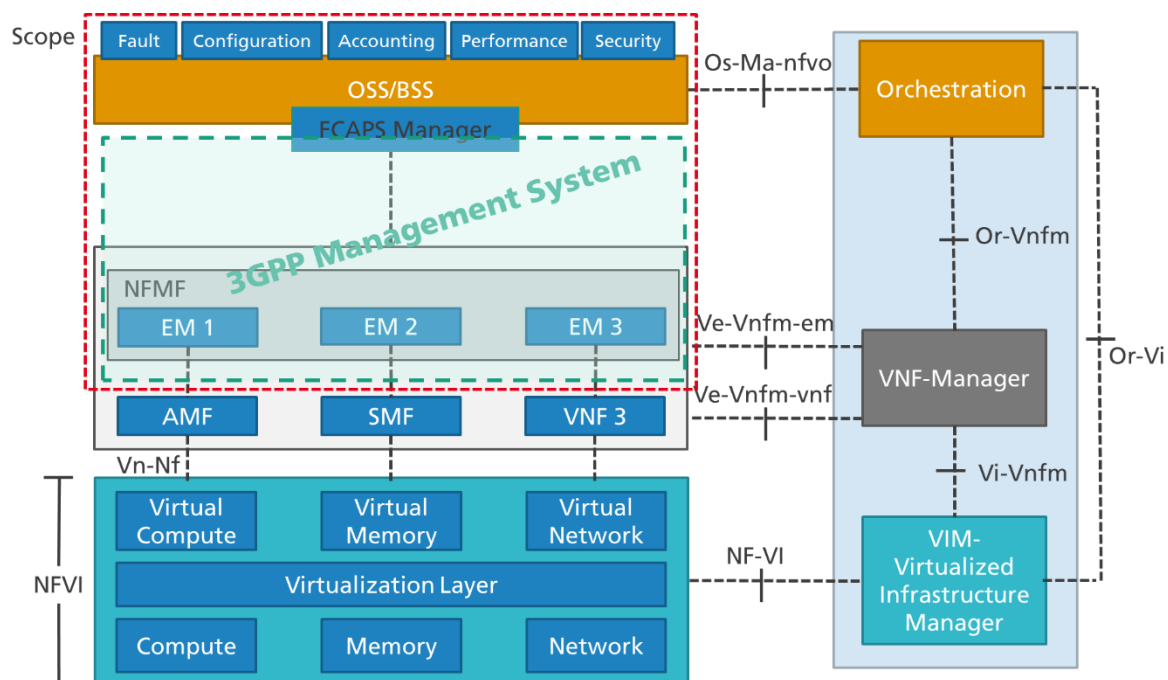


Figure 3 3GPP Management System

4.1.3 Scope

This section defines the scope of the term "5G management" in regard to which parts of the 5G system shall be part of the Detailed Risk Analysis. The relevant part is outlined in red in the architecture diagram.

For the task of configuring and monitoring functions, the International Organization for Standardization (ISO) has standardized the **FCAPS** management framework for networks. FCAPS categorizes the specific objectives of network management into five levels. These five levels are Fault Management (F), Configuration (C), Accounting (A), Performance (P) and Security (S).

FCAPS management forms the basis of 5G management. It is integrated into the element management functions and realizes the connection to the OSS. OSS stands for Operations Support System and it is responsible for the operational management of the entire 5G network. The OSS is also subdivided according to the FCAPS principle and offers its own function modules for fault management, configuration, accounting, performance, and security, in addition to other management functions. The element management functions implement the FCAPS management of the various network functions within 5G core network, and of functions in the radio access network, at different levels of granularity. A network function can have its own element management function. Alternatively, an element management function can take over the management of several network functions in a 5G core network. In 5G management, for example, this task is performed by the **NFMF**, the NF Management Function.

The new modular service-based architecture (SBA) of the Standalone 5G core network, which has many small function blocks, requires a new concept that modularizes the interfaces between the core network and OSS. In core networks of earlier mobile radio standards, many of these interfaces were file-based. With the TS 28.533 standard, the 3GPP specifies a new interface concept in the form of the Service Based Management Architecture (**SBMA**). The main task of the architecture's implementation is the management of network slices and network functions in the core network and RAN. It should be noted that not all management tasks have been converted to this architecture. The 3GPP is continuously developing the system with each new release. Appropriate standards already exist for performance and fault management. The TS 28.552 standard defines various performance measurement methods in the different network variables and in the radio access network. In addition, the TS 28.550 standard defines the associated management services for 5G networks, including network slicing. Fault management based on SBMA is defined in 3GPP standard TS 28.111.

Further 5G management modules are briefly outlined below. The TS 28.540 standard defines the 5G Network Resource Model (**NRM**). It is used to support the management of a variety of 5G radio access network functions and 5G core network functions. Furthermore, the 3GPP has recognized the problem that the fine-grained modular structure of the 5G core network, the NG-RAN, and the introduction of network slices have led to a system complexity that can no longer be managed by human administrators alone. For this reason, the 3GPP relies on "self-organizing networks" (**SON**) and management data analytics (**MDA**) in the context of orchestration, management, and automation (**OMA**). The MDAS (Management Data Analytics Service), which is responsible for the processing of historical data and machine learning support, is standardized within TS 28.104. Its services form an important basis for the correct administration and management of a 5G network.

The functions that are particularly relevant for the management of the 5G core network, the radio access network, and also multi-access edge computing are summarized below. A successful attack on these functionalities can have a massive impact on the availability and integrity of data and services and on the confidentiality of the 5G system.

- Operations Support System (OSS)
- Element Management Functions (EMF)
- NF Management Function (NFMF)
- Software Based Management Architecture (SBMA)
- Fault Management
- Configuration Management
- Accounting Management
- Performance Management
- Security Management
- Network Slicing Management
- 5G Network Resource Model (NRM)
- Orchestration, management and automation (OMA)
- Self-organizing networks (SON)
- Management Data Analytics (MDA)

5 Risk Scenarios

This chapter provides the current view on a range of risk scenarios. It deals with an estimation of probabilities that each of these risk scenarios could become reality. These estimations are based on detailed discussions of potential scenarios of weaknesses or attacks. The discussions conclude with an assessment of the probability of occurrence.

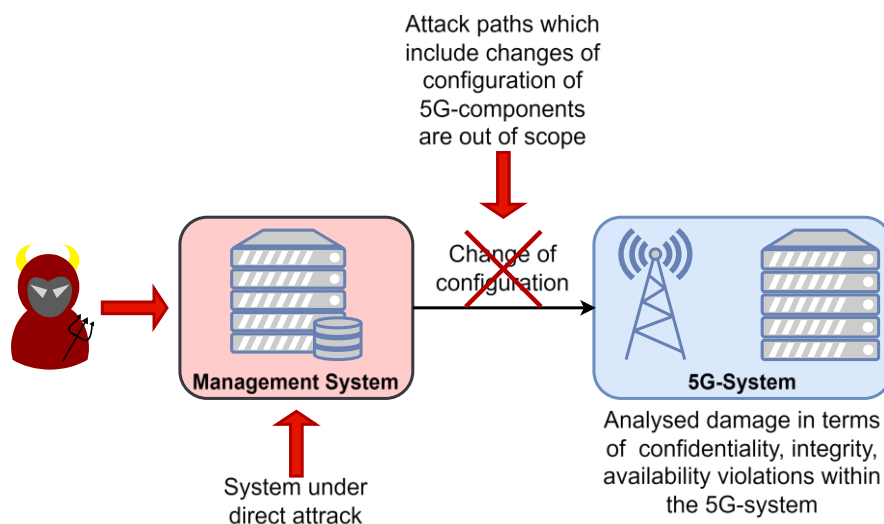


Figure 4: General risk settings analysed in this study

Only the management components are considered for reasoning about potential attack paths, i.e. only attacks which are based on threats and vulnerabilities within the management system are considered. Regarding the potential damage, only the 5G system as such is considered but not the management system itself. E.g. confidentiality refers to the protection of the user plane or control plane data of the 5G system – but not the confidentiality of management system related data. And finally and most notable, attacks which are based on changing or manipulating the configuration of the 5G components itself (with the help of the compromised management system) are out of scope since this kind of attacks are discussed in the detailed risk analyses of the subsystems RAN, Core, MEC, Transport, Infrastructure, and IMS/Interconnect.

The following table shows if the Management layer is a relevant subsystem of each risk scenario. The risk scenarios themselves are further described in the framework document.

Perspective	Section	Risk Scenario	Management	CIA
Public MNO Perspective	5.2.1	Emergency Call (TKG § 164)	x	A
	5.2.2	Cell Broadcast (TKG § 164a)	x	A
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	x	C
	5.2.4	Emergency Preparedness (TKG § 184)	x	A
	5.2.5	Regional Network Failure	x	A
	5.2.6	Multi-Regional Network Failure	x	A
	5.2.7	Failure of Mobile Service	x	A
	5.2.8	Fraud	x	I
	5.2.9	Subscriber Data Leak	x	C
User Perspective	5.3.1	Urgent Telephone Calls Not Possible	x	A
	5.3.2	Authenticity of the Receiver of a Call	x	I
	5.3.3	Integrity of Transmitted Data		I
	5.3.4	User's Identity	x	I
	5.3.5	Eavesdropping of a Communication	x	C
	5.3.6	Slowing Down/Restricting the Connection	x	A
	5.3.7	Connection Failure	x	A
	5.3.8	Tapping or Altering of the Geographical Position	x	C
Private 5G Network Operator Perspective	5.4.1	Network Failure	x	A
	5.4.2	Degraded Services/Connections	x	A
	5.4.3	Information Disclosure		C
	5.4.4	User Identities (Authenticity/ Integrity)	x	I

5.1 Attack Paths and Risk Sources

For each risk scenario a couple of different attack paths and risk sources will be discussed within this document. The result of the discussion will be a probability level for the impact defined in the respective risk scenario.

The term “attack path” refers to a sequence of steps or actions that an attacker can take to exploit vulnerabilities in a system or network, leading to security issues. It outlines the route or method an attacker follows to achieve his goals, often involving a series of interconnected actions that exploit weaknesses in a system's defences.

A “risk source” is defined as the origin or cause of a potential risk or threat. A risk source is identified as the factor or element that can lead to vulnerabilities or hazards.

As many attack paths or sub-sequences of attack paths and risk sources are similar for most of the risk scenarios, each attack path and risk source will be introduced from a management perspective in this chapter.

In general, an attacker with legitimate access to network operator's assets is called “internal attacker”. An internal attacker could have access to physical or cryptographic keys, to buildings or rooms or to networks or systems.

5.1.1 Misconfiguration (AP1)

Misconfigurations are risk sources which concern all configurable parts of a system. A legitimate configuration can either be deployed by employees of the MNO or the manufacturer or by configuration tools. Therefore, causes of misconfiguration can be either human errors or software bugs thus happening unintentionally. On the other hand, misconfigurations can also be intentionally used as part of an attack path.

Misconfigurations may persist undetected if the impacted functionality is not used regularly or if configuration or security impacts of that functionality are not monitored.

It is important to mention that many attack paths using misconfigurations can affect all protection goals. E.g., integrity related issues could often lead to confidentiality issues.

Misconfiguration w.r.t. distributed systems is one of the most often reported risk source, e.g., it is part of the Open Web Application Security Project (OWASP) Top Ten list [OWASP] of web application security risks since 2010. According to the 2024 edition of the “The state of Kubernetes security report” [REDHAT] by Red Hat, 40% of the 600 responses to an online survey among DevOps, engineering, and security professionals (companies with more than 100 employees) answered, that they have faced security misconfiguration within the last 12 months. Moreover, 27% of respondents see misconfigurations as the risk they are most worried about.

5.1.1.1 Confidentiality (AP1-C)

(1) Misconfiguration of MGMT system protocols

The management system most likely provides some kind of UI or API. If this web UI or API can be accessed via HTTP instead of HTTPS, sensitive data could be exposed. Apart from the example concerning HTTP and HTTPS, there are many other ways to render the connection between the administrator and the management interface insecure. Another example could be the use of insecure cryptography [SSTLS][E].

(2) Misconfiguration of Endpoints

Data endpoints in the management system are functions and interfaces, mostly exposed in the form of APIs, that create and send reports on a regular basis or just offer access to different information. E.g., reports could be sent in the form of an email. If the recipients' emails are misconfigured causing the report being sent to an incorrect recipient, sensitive data could be leaked to third parties [MICRO] [MAIL] [TELEAPI] [E].

(3) Misconfiguration of security controls

The configuration of individual settings and parameters that are set by default or may be allowed as an insecure option can be overlooked or incorrectly configured. This may lead to the usage of insecure configurations. The following list shows examples of the feasibility and the impact of misconfigurations:

- AS Security allows to deactivate encryption (TS 38.331 5.3.1.2 v18.1.0). If this is the default case, it is possible that messages are sent unencrypted and can be intercepted by an attacker [TS38331]. (RAN)
- NAS Security allows to deactivate encryption (TS 24.501 4.4.5 v18.6.0). If this is the default case, it is possible that messages are sent unencrypted and can be intercepted by an attacker [TS24501]. (Core)
- Insecure default values for the configuration of IPSec, which is used for example for the N2 interface (TS 33.501 9.2 v16.17.0) [TS33501]. (Transport)

Creating an exhaustive list of possible misconfigurations would go beyond the scope of this document because considerations about misconfiguration of the different 5G components itself is part of the detailed risk analyses of the other subsystems.

Probability Assessment

The probability of misconfiguration leading to breaking confidentiality of a MGMT system is considered **MEDIUM**. First of all, there is data in the MGMT system which is confidential, so misconfigurations as described above, could leak this sensitive data. Since the available data is highly confidential, mobile operators are expected to have security measures in place to prevent breaches. On the other hand, the probability of attacks and risks increases because of the multitude of possible misconfigurations and endpoints that exist in such a complex system, such as in 5G. Therefore, the combination of the high probability of misconfigurations in general as described here and in more detail in the preface combined with security measures and best-practices of MNOs leads to the overall probability being assessed as **MEDIUM**.

5.1.1.2 Integrity (AP1-I)

(1) Misconfiguration of MGMT system protocols

The management system must provide some kind of UI or API. If this web UI or API can be accessed via HTTP instead of HTTPS, sensitive data could be exposed. Apart from the example concerning HTTP and HTTPS, there are many other ways to render the connection between the administrator and the management interface insecure. Another example could be the use of insecure cryptography [SSTLS][E].

(2) Misconfiguration of the user administration

Web UIs or APIs may have preconfigured default users with known credentials creating a common attack path. Incorrect configuration of user groups' rights and privileges can pose another attack path. For example, users could make changes to the system that normally require higher access rights [NISSAN]. On top of that, attack paths can include shared user accounts or long-lived credentials stemming from the absence of an active user administrator [E].

(3) Automatic data processing

Data must be processed and prepared for presenting the relevant information in the MGMT system and for analysis reports. If algorithms and procedures are configured incorrectly, the integrity of the data as it is viewed by users may be violated [E].

Probability Assessment

The probability of misconfiguration leading to issues with MGMT system's integrity is considered **MEDIUM**. Here the same reasoning as for confidentiality can be applied. This is due to the fact that misconfigurations affecting integrity are slightly less frequent than affecting the confidentiality. With the remaining attack paths being quite similar, the overall risk is also considered **MEDIUM**.

5.1.1.3 Availability (AP1-A)

(1) Incorrect energy saving settings

Saving energy is an ever-growing topic. System optimization can save money and can have a positive impact on the environment and an operator's reputation. However, wrong or too strict configurations of power management can lead to poor performance or outages of a system [E].

(2) Incorrect logging configuration

If the 5G management's logging system is configured badly different problems can occur, as it is a critical part to make issues observable and consequently maintain the system's availability. However, excessive logging might lead to system's unavailability as memory space might be limited to store the logging data [SYSLOG]. The system might run into a situation where there is no memory left. This might lead to incomplete logging data or affecting the system in such a way that the availability of the system might be in danger, as no more memory can be accessed [E].

(3) No password recovery for management system

If there is no password recovery in place and the login for the management system is lost, normal users or administrators might lose access to the system which can lead to a situation where the MGMT becomes unavailable. Therefore, information or settings could not be entered in the MGMT system anymore and the availability could not be guaranteed. In the same sense intentional misconfigurations of credentials are possible to lock out other users [E].

(4) Badly configured login for management system

Having configured a login guard, an account could be suspended for multiple hours if you repeatedly enter a wrong password. An attacker who is aware of login names might use this to intentionally suspend accounts. Users or administrators would not be able to access the system anymore which might lead to a situation where settings could not be applied to the system, putting the availability of the system at risk [E].

(5) Wrong or insufficient thresholds for warnings

An important function of a management system is to observe the system state and send warnings if problems occur.

In a situation of error, if no warning is sent because the threshold is set too high the system might run into problems.

If the threshold is too low, many warnings might be sent. The administrators might not be able to handle all of them or might miss important ones. This might lead to a situation where important warnings are lost which might endanger the availability of the system [E].

(6) Wrongly configured APIs

This attack path refers to the application layer. If the APIs or web UIs connecting the management system and the other 5G systems are not configured correctly, there might be a situation where changes made in the management system are not transmitted to the 5G sub-systems. Additionally, changes made for one parameter might affect a completely different parameter [E].

(7) Wrong IP addresses and routing tables

This attack path refers to the network layer. Wrong routing/addressing might lead to a situation where parts of the management system cannot communicate with its managed components or each other [E].

(8) Insufficiently configured load balancing

In networks load balancing is an important topic. If configurations in this domain are not precisely made the availability of the system might be affected [E] [GOOCL].

(9) Insufficiently configured redundancy mechanisms

Redundancy is important for critical systems, because guaranteeing that a given component will never fail is impossible. If redundancy mechanisms are configured insufficiently problems can occur if failing components have no fallback or redundant component implemented. A somewhat opposite effect can occur if unnecessarily (a large amount of) redundant components are operated. The latter will have negative impact on resource consumption which could eventually lead to unavailable resources. Both cases might lead to availability problems of the system [E].

(10) Misconfigured certificate management

If certificate management is misconfigured, situations might occur where MGMT components cannot communicate with each other and consequently rendering them unavailable. An example would be that certificates expire and have not been renewed in time [E][SECTIGO]. This can become especially problematic, if additional security mechanisms are in place such as HTTP Strict-Transport-Security (HSTS) and the components of the management system do not allow to override the HSTS policy. Certificate issues may lead to situations, in which the certificate of the certificate management component becomes outdated and therefore no secure management connection should be established, which would be necessary to renew the outdated certificates [E].

Probability Assessment

The probability of misconfiguration leading to issues with the MGMT system's availability is considered **HIGH**. This is due to the numerous attack paths that demonstrate how various misconfigurations—whether accidental or intentional—can affect the system's availability. Furthermore, as mentioned in the preface, the overall risk of misconfigurations leading to issues within a given system is also **HIGH**. The combination of those aspects yields a **HIGH** probability for issues with availability due to misconfigurations in the 5G MGMT domain.

5.1.2 Insufficient Access Controls (AP2)

Insufficient physical or remote access controls can empower internal and external attackers to have access to 5G management interfaces and therefore to sensitive monitoring, user and administrative information. Insufficient access control is a risk source that can enable attackers to access management systems (either physically, remotely or through human intermediaries). Due to the critical nature of insufficient access control all security goals might be at risk.

Within this assessment it is worth to note, that an attacker who has gained access can change configurations of underlying components in the management systems. Since this is not part of this risk analysis, it will not have any impact on the resulting risks.

5.1.2.1 Confidentiality (AP2-C)

(1) Exploitation of weak authentication and access control

This attack path can potentially compromise all security goals by giving access to the configuration of the MGMT system or to information within the management domain. Manipulating the MGMT system (OSS/BSS) by having direct access is considered to be integrity related [E] (see 5.1.2.2 for further details). However, having general access to information like user data within the management is a confidentiality concern.

(2) Credential theft / identity abuse

This attack path can potentially compromise all security goals by providing access to the whole configuration of the MGMT system or to information within the management domain. Manipulating the MGMT system by having direct access is considered to be integrity related [E] (see 5.1.2.2 for further details). However, having general access to information like user data within the management is a confidentiality concern.

(3) Injection of malicious software (malware)

An attacker might inject malicious software such as backdoors into supply chains or in open-source software which is used in the management system enabling eavesdropping [CYSEC] [GTPDOOR] [SOLAR] [HUA] [LOG4SH] [XZ].

(4) Injection of malicious hardware (hardware trojan)

Malicious injections of hardware for example in FPGAs or ASICs during any process in the supply chain or even the design phase may open backdoors, cause faults, enable eavesdropping of keys or load evil software at run-time. Another example are random number generators which might be mitigated to produce weak random numbers. In general, this can lead to eavesdropping or tampering of communication in the MGMT system [NSA] [HUA] [EDSNOW] [HTROJ].

(5) Insecure APIs to and from management systems

If APIs are not sufficiently secured by authentication, adversaries may exploit MGMT functionality by Man-in-the-Middle attacks, spoofing or eavesdropping, potentially affecting all protection goals in the MGMT plane [SUPATK].

(6) Privilege escalation

Extensive default privileges or no privilege management in edge or cloud services, or such services running with excessive privilege may allow access of malicious users or containers to confidential data or MGMT functionality, e.g., to create new users or containers, to change passwords or manipulate log data. The MGMT system could also be shut down resulting in unavailability [ESEC] [SUPATK] [PETYA].

Probability Assessment

The probability of insufficient access control leading to breaking confidentiality of a MGMT system is evaluated as **MEDIUM**. In general, there is no large user base with access to the MGMT systems, but only a few administrators. This eases access and privilege control by a wide margin. Furthermore, access to MGMT systems is mostly granted to a few interfaces which simplifies their configuration and monitoring. Due to the MNOs' deployment of well developed software from trusted vendors, a supply chain [NSA] [HUA] or malicious backdoor injection [SOLAR] in MGMT systems is less likely. The most likely incidents are assumed to relate to insecure APIs [OWASP] which might enable the eavesdropping of information of the MGMT system [SUPATK].

5.1.2.2 Integrity (AP2-I)

(1) Exploitation of weak authentication and access control

Under this category fall weak passwords, weak or no enforcement of multi factor authentication (MFA), usage of default passwords (e.g. rainbow list passwords), long-lived credentials, insecure public key infrastructure usage and/or algorithms (e.g. backwards compatibility of old protocols, out of date certificates, insecure protocols), insufficient login configuration of login shell enabling brute force and other attacks, physical access through direct connections, interceptions, resets, etc. (e.g. no plug-and-play control, backup protection) or leaking of logging information or access to databases [SECENG] [SECSPEC] [ATNT24] [KYIV] [STUX].

(2) Credential theft / Identity Abuse

Attack paths include phishing of user credentials, spoofing of legitimate users, placing key loggers or other sniffing hardware on management devices or hacking devices used for accessing the management. In a similar sense, access can be gained through unlocked or connected devices with saved password information (stealing, etc.) [ENISATHREAT] [ENISAINC] [ATNT24].

(3) Injection of malicious software (malware)

An attacker might inject malicious software such as backdoors into supply chains or in open-source software which is used in the management system enabling manipulation of these systems [CYSEC] [GTPDOOR] [SOLAR] [HUA] [LOG4SH] [XZ].

(4) Injection of malicious hardware (hardware trojan)

Malicious injections of hardware for example in FPGAs or ASICs during any process in the supply chain or even the design phase may open backdoors, cause faults, enable eavesdropping of keys or load evil software at run-time. Another example are random number generators which might be mitigated to produce weak random numbers. In general, this can lead to eavesdropping or tampering of communication in the MGMT system [NSA] [HUA] [EDSNOW] [HTROJ].

(5) Insecure APIs to and from management systems

If APIs are not sufficiently secured by authentication, adversaries may exploit MGMT functionality by Man-in-the-Middle attacks, spoofing or eavesdropping, potentially affecting all protection goals in the MGMT plane [SUPATK].

(6) Privilege escalation

Extensive default privileges [ESEC] or no privilege management in edge or cloud services, or such services running with excessive privilege may allow access or manipulation of containers, virtual machines or MGMT functionalities, e.g., to create new users or power off containers, mount wrongful data volumes into virtual machines, change passwords or manipulate log data. The most common attacks are ransomware [PETYA] or similar attacks that either steal data and then delete the existing data or cause the MGMT and underlyings systems to be unavailable, e.g., by user permission tampering or encryption.

Probability Assessment

The probability of insufficient access control leading to breaking the integrity of a MGMT system is evaluated as **MEDIUM**. In general, there is no large user base with access to the MGMT systems, but only a few administrators. This eases access and privilege control by a wide margin. Furthermore, access to MGMT systems is mostly granted only to a few interfaces which simplifies their configuration and monitoring. Due to the inhouse development of many MGMT systems of MNOs, a supply chain [NSA] [HUA] or malicious backdoor injection [SOLAR] is less likely. Furthermore, in contrast to Confidentiality (AP1-C), insufficient access control of APIs is less likely to allow for direct control of the MGMT systems. Even so, the probability of an attacker gaining access is still considered to be **MEDIUM**.

5.1.2.3 Availability (AP2-A)

In general, availability is affected, if an attacker has the ability to cause the MGMT system to be unavailable through tampering of user permissions or other attack paths, see section 5.1.2.2 above. In other words, the risk of unavailability due to insufficient access control stem from the fact that integrity protection has failed in most cases. So while the attack paths described below are not the risk source, they still are the underlying causes that lead to unavailability of the MGMT systems.

(1) Exploitation of weak authentication and access control

In this category belong weak passwords, weak or no enforcement of multi factor authentication (MFA), usage of default passwords (e.g. rainbow list passwords), long-lived credentials, insecure public key infrastructure usage and/or algorithms (e.g. backwards compatibility of old protocols, out of date certificates, insecure protocols), insufficient login configuration of login shell enabling brute force and other attacks, physical access through direct connections, interceptions, resets, etc. (e.g. no plug-and-play control, backup protection) or leaking of logging information or access to databases [SECENG] [SECSPEC] [ATNT24] [KYIV] [STUX].

(2) Credential theft / Identity Abuse

Attack paths include phishing of user credentials, spoofing of legitimate users, placing key loggers or other sniffing hardware on management devices or hacking devices used for accessing the management. In a similar sense, access can be gained through unlocked or connected devices with saved password information (stealing, etc.). [ENISATHREAT] [ENISAINC] [ATNT24].

(3) Injection of malicious software (malware)

An attacker might inject malicious software such as backdoors into supply chains or in open-source software which is used in the management system enabling manipulation of these systems [CYSEC] [GTPDOOR] [SOLAR] [HUA] [LOG4SH] [XZ].

(4) Injection of malicious hardware (hardware trojan)

Malicious injections of hardware for example in FPGAs or ASICs during any process in the supply chain or even the design phase may open backdoors, cause faults, enable eavesdropping of keys or load evil software at run-time. Another example are random number generators which might be mitigated to produce weak random numbers. In general, this can lead to eavesdropping or tampering of communication in the MGMT system [NSA] [HUA] [EDSNOW] [HTROJ].

(5) Insecure APIs to and from management systems

If APIs are not sufficiently secured by authentication, adversaries may exploit MGMT functionality by Man-in-the-Middle attacks, spoofing or eavesdropping, potentially affecting all protection goals in the MGMT plane [SUPATK].

(6) Privilege escalation

Extensive default privileges [ESEC] or no privilege management in edge or cloud services, or such services running with excessive privilege may allow access or manipulation of containers, virtual machines or MGMT functionalities, e.g., to create new users or power off containers, mount wrongful data volumes into virtual machines, change passwords or manipulate log data. The most common attacks are ransomware [PETYA] or similar attacks that either steal as much data as possible and then delete existing data or cause the MGMT and underlyings systems to be unavailable, e.g. through user permission tampering or encryption.

Probability Assessment

The probability of insufficient access control leading to breaking the availability of a MGMT system is evaluated as **MEDIUM**. This is based on the preface of this section (5.1.2.3) that availability itself is only a consequence of failed integrity due to insufficient access control. Generally, once someone has gained any

form of access or control of a MGMT system it is easy to cause availability issues, unless the system is specifically built around this specific security goal which is not the case for most MGMT systems today [E]. Consequently, the probability to cause unavailability in a MGMT system is rated as high as the one for integrity, see 5.1.2.2.

5.1.3 Implementation Flaw (AP3)

Implementaton Flaws are coding errors which are still persistent in the executed program in a productive system after insufficient testing or human error. Implementation flaws occur especially in low level programming languages like C or C++ where developers must take care of memory management themselves as no garbage collection is available as in languages like Python or Go; or borrow checking as in Rust. This can lead to memory related errors such as “use-after-free”, “index-out-of-bounds” or “stack overflows”, which are the most common error source in programming [NSAMEMORY]. Implementation flaws are often the root cause for other attack paths. For example, access control systems can be broken due to implementation flaws, making it possible for an attacker to gain unauthorized access to a system. Implementation flaws might also lead to outages of the system as errors might not be handled correctly leading to an ungraceful failure of a program or a complete system.

5.1.3.1 Confidentiality (AP3-C)

(1) Classical implementation errors

Numerous mistakes can be made when developing the management system. If the system is developed as a web application, the most common errors are listed in the OWASP top 10. However, even if the system is not developed as a direct web application, the errors listed there should be considered as such implementation errors can lead to data leakage throughout any system [OWASP] [ATHEN].

(2) Outdated third-party software

Numerous third-party software might be used when implementing a management system. Outdated and insecure software versions might be used [HEARBL] [LOG4SH] [WANCERY] [PETYA] [EQUIFAX] [DIRCOW] [TAGSTOE].

(3) New weaknesses in third-party software

Numerous third-party software might be used when implementing a management system. Even if the protocols used are considered secure, it cannot be ruled out that they contain vulnerabilities that will sooner or later be discovered and exploited [HEARBL] [LOG4SH] [WANCERY] [PETYA] [EQUIFAX] [DIRCOW] [TAGSTOE].

(4) Supply chain attacks

In addition to unintentional software errors, deliberate attacks can also be a problem. For example, supply chain attacks can be used to intentionally introduce vulnerabilities into systems and software [SOLAR].

Probability Assessment

The probability of implementation flaws leading to a loss of confidentiality of a MGMT system is evaluated as **MEDIUM**. There are many examples where implementation flaws lead to confidentiality issues, as described above. However, since the data in the MGMT system is quite critical, and leakages could lead to big problems for the mobile telecommunication provider, the expected effort to secure the data is also high. Therefore, there is high effort in protecting the data but also strong evidence, that also well developed and critical applications can contain vulnerabilities, which in combination leads to a probability of **MEDIUM**.

5.1.3.2 Integrity (AP3-I)

(1) Classical implementation errors

Numerous mistakes can be made when developing the management system. If the system is developed as a web application, the most common errors are listed in the OWASP top 10. However, even if the system is not developed as a direct web application, the errors listed there should be considered [OWASP] [ATHEN].

(2) Outdated third-party software

Numerous third-party software might be used when implementing a management system. Outdated and insecure software versions might be used [HEARBL] [LOG4SH] [WANCERY] [PETYA] [EQUIFAX] [DIRCOW] [TAGSTOE].

(3) New weaknesses in third-party software

Numerous third-party software might be used when implementing a management system. Even if the protocols used are considered secure, it cannot be ruled out that they contain vulnerabilities that will sooner or later be discovered and exploited [HEARBL] [LOG4SH] [WANCERY] [PETYA] [EQUIFAX] [DIRCOW] [TAGSTOE].

(4) Supply chain attacks

In addition to unintentional software errors, deliberate attacks can also be a problem. For example, supply chain attacks can be used to intentionally introduce vulnerabilities into systems and software [SOLAR].

(5) Integrity violation through implementation errors

The integrity of the management system can be affected by classical implementation errors such as race conditions, off-by-one errors, buffer or stack overflow [E] [NSAMEMORY].

Probability Assessment

The probability of implementation flaws leading to problems with the integrity of a MGMT system is evaluated as **MEDIUM**. Here the same reasoning as for the confidentiality case can be applied. We also have a necessity of high efforts from MNOs to ensure the system's integrity is secured. This is due to possible problems with billing or contracts, which are the main income sources of the MNOs. Also, again, as shown above, we have strong evidence that implementation flaws leading to attack paths on integrity can never be fully mitigated. Therefore, we also consider the risk as **MEDIUM** here.

5.1.3.3 Availability (AP3-A)

(1) System failure through implementation errors

The availability of the management system can be affected by classical implementation errors that may cause a system failure often due to buffer overflows, null-pointer exceptions, deadlocks or infinite loops [ARIANE][E][CROWDS].

(2) Performance decrease through implementation errors

The availability of the management system can be affected by classical implementation errors that may cause a system performance decrease, e.g., memory leaks, unnecessary locks, unnecessary resource usage or resource starvation [E].

Probability Assessment

The probability of implementation flaws leading to problems with the availability of a MGMT system is evaluated as **HIGH**. In principle we consider the risk of availability problems due to implementation flaws as high for every existing system. This is because there are many examples of implementation flaws causing availability problems, even in critical and highly tested systems like [ARIANE][E].

5.1.4 Specification Weakness (AP4)

To the authors' knowledge, no publicly available specification for a 5G MGMT system exists. Therefore, operators might specify and implement their own MGMT solutions, which are not available to the public.

5.1.4.1 Confidentiality (AP4-C)

(1) Missing or insufficient specification of protocols between MGMT components

A missing or insufficient specification of protocols between MGMT components and managed components may lead to data leakage. If no adequate authentication or encryption (e.g. TLS, OAuth2.0) of the management interfaces is specified, data or management calls can be observed and altered, leading to confidentiality, integrity and availability issues [ESEC] [GTPDOOR].

(2) Specification weaknesses in protocols used by a MGMT system

TLS, HTTP or other protocols may contain weaknesses due to their specification which can lead to issues with all three security goals [ESEC] [GTPDOOR].

(3) Exploitable specification of Lawful Interception (LI)

A system that intercepts communication can pose an entry point for further attacks. Specification weaknesses could lead to exploitation of such a system which could lead to unallowed usage and observation of UEs [ATHEN].

Probability Assessment

The probability of specification weaknesses leading to mitigating the confidentiality of a MGMT system is evaluated as **LOW**. It is assumed that state-of-the-art security protocols like TLS or OAuth 2.0 are used which have existed for decades by now and therefore have been tested and improved continuously by the global community. Even though zero-day exploits sometimes appear [HEARBL] those are usually fixed quickly. Even then, prior network information is still needed to carry out effective attacks with a zero-day exploit at hand. In the same sense LI has existed for a long time and has improved its security since the Athen's affair [ATHEN]. The missing specification of the MGMT system and the used protocols may have a negative impact on the security of the MGMT system, however, the probability of this leading to a successful attack is very hard to quantify and expected to be rather low. In summary, the probability is evaluated under the assumption that systems are kept up-to-date to the latest IT-security standards.

5.1.4.2 Integrity (AP4-I)

(1) Missing or insufficient specification of protocols between MGMT components

A missing or insufficient specification of protocols between MGMT components and managed components may lead to data leakage. If no adequate authentication or encryption (e.g. TLS, OAuth2.0) of the management interfaces is specified data or management calls can be observed and altered, leading to confidentiality, integrity and availability issues [ESEC] [GTPDOOR].

(2) Missing or Specification weaknesses in protocols used by a MGMT system

TLS, HTTP or other protocols may contain weaknesses due to their specification which can lead to issues with all three security goals [ESEC] [GTPDOOR].

Probability Assessment

The probability of specification weaknesses leading to mitigating the integrity of a MGMT system is evaluated as **LOW**. As described in section 5.1.4.1, it is expected that potential weaknesses within established protocols have a relatively low impact on the integrity of 5G MGMT. In a similar sense the use of up-to-date security protocols is assumed in order to justify this low probability assessment. The impact of missing specifications is assumed low because MNOs deploy commercial MGMT software and therefore its code and

weaknesses are not publicly available, further reducing the possibility of misuse. On top of that, such software will be audited and tested thoroughly by the corresponding suppliers of the MGMT software.

5.1.4.3 Availability (AP4-A)

(1) Missing or insufficient specification of protocols between MGMT components

A missing or insufficient specification of protocols between MGMT components and managed components may lead to data leakage. If no adequate authentication or encryption (e.g. TLS, OAuth2.0) of the management interfaces is specified, data or management calls can be observed and altered, leading to confidentiality, integrity and availability issues [ESEC] [GTPDOOR].

(2) Denial of Service (DoS) due to Specification weaknesses

A fault in the specification could enable DoS attacks on the MGMT system if interfaces of the system are accessible from outside of a secured perimeter [DOSCONT].

Probability Assessment

The probability of specification weaknesses leading to mitigating the availability of a MGMT system is evaluated as **LOW**. Similarly to the assessment in sections 5.1.4.1 and 5.1.4.2, it is expected that potential weaknesses within established protocols have a relatively low impact on the availability of 5G MGMT. Additionally, the use of up-to-date security protocols is assumed in order to justify this low probability assessment. The impact of missing specifications is assumed to be low. A detailed analysis can not be conducted because MNOs deploy commercial software of MGMT systems from mobile network vendors whose code and weaknesses are not available to the public. On top of that such software is expected to be audited and tested thoroughly by the suppliers of the MGMT software.

5.1.5 Failure of Power Supply (AP5)

Failure of power supply means that the considered system loses its power supply. This may happen for example by a terrorist attack, a natural disasters or by aging processes.

5.1.5.1 Confidentiality (AP5-C)

(1) Low security emergency mode

It is very unlikely that a failure of power supply will lead to problems with the MGMT's confidentiality or integrity. A possible path of failure would be that the system switches to an emergency mode in the event of problems with the power supply. In this mode security mechanisms might be restricted to save energy and promote availability. One example could be that the management interface is only made available via HTTP instead of HTTPS [E].

Probability Assessment

Breaches of confidentiality due to failure of power supply are very unlikely. A low security emergency mode may be the only attack path that could potentially lead to confidentiality breaches [E]. Therefore, the probability of a failure of power supply leading to mitigating the confidentiality of a MGMT system is evaluated as **LOW**.

5.1.5.2 Integrity (AP5-I)

(1) Write of data

If power is lost during a write of data, it might happen that the data being written is lost or corrupted [E].

Probability Assessment

The probability of a failure of power supply leading to mitigating the integrity of a MGMT system is evaluated as **LOW**. The existence of low security emergency modes in MGMT systems and their impact on them is unlikely since security systems do neither use a lot of energy compared to the overall system nor does the small number of management data warrants a change in encryption. The scenario of data corruption due to write errors rooted in power outages seem to happen very rarely since power outages itself are already very unlikely. Even so, it would be even more unlikely that faulty write of data affects control data in such a way that it would cause integrity damage. Furthermore, a control command is not executed if the command is corrupted randomly, instead it would need to get corrupted into another legitimate control command. On top of this the corruption has to go unnoticed after a restart. Additionally, many modern file systems (e.g. journaling file systems) or database systems are designed with possible power outages in mind. Therefore, they provide measures to prevent integrity issues, like redundant and distributed systems or error correction for memory. Moreover, it can be assumed, that the related systems are operated using means of backup power supply, e.g. uninterruptible power supply (UPS).

5.1.5.3 Availability (AP5-A)

(1) Failure of power supply by accident or catastrophic events (disasters)

Natural disasters such as flooding, earthquakes, lightnings or fires can interrupt or jeopardize the supply of energy. This can especially lead to issues in availability of the MGMT system or the complete network. [AMA14] [AMA16] [GOO15] [OVH21].

(2) Intentional manipulation of power supply that leads to its failure and service unavailability

Terrorism, sabotage or fraud can lead to destruction of the power supply necessary to operate MGMT systems. With physical or even remote access to the power supply it could intentionally be destroyed leading to outage of the MGMT system or other parts of the network [ENISATHREAT] [STUXNET] [HUA].

(3) Missing redundancy of power supply implementation

A complete lack of, or insufficient redundancy concepts may cause key MGMT systems to be unavailable in case of power supply failure [ENISAINC] [E].

Probability Assessment

The probability of a failure of power supply leading to mitigating the availability of a MGMT system is evaluated as **LOW**. The probability of failure of power supplies due to disasters [AMA14] [AMA16] [GOO15][OVH24] is assumed to be LOW because in Germany the occurrence of extreme disasters is relatively seldom. A disaster that makes the power supply of the MGMT system fail is even more unlikely. The probability of an attacker targeting the relevant power supplies also often requires physical access, sufficient time, and makes a discovery of identity likely. In summary we estimate this to be difficult and therefore unlikely. Also, please note that we do not consider wars, or large terroristic szenarios.

5.1.6 Failure of Hardware Components or Unavailability of Spare Parts (AP6)

Failure of MGMT hardware will not lead to immediate outage of the cellular system, however, unavailability of the MGMT system over extended periods of time will lead to an accumulation of unresolved errors or security incidents, that might not be recognized. This increases the possibility of a system failure.

5.1.6.1 Confidentiality (AP6-C)

No relevant attack path concerning confidentiality could be identified.

5.1.6.2 Integrity (AP6-I)

No relevant attack path concerning confidentiality could be identified.

5.1.6.3 Availability (AP6-A)

(1) Failure of hardware by accident or catastrophic events (disasters)

Natural disasters such as flooding, earthquakes, lightnings or fires can destroy facilities that are essential for the operation of an 5G MGMT system. This can especially lead to issues in availability of the MGMT system or the complete network [ENISATHREAT] [ENISAINC].

(2) Intentional Manipulation of hardware that leads to its failure and service unavailability

Terrorism, sabotage or fraud can lead to destruction of hardware necessary to operate MGMT systems. With physical or even remote access to hardware these could intentionally be destroyed leading to outage of the MGMT system or other parts of the network [ENISATHREAT] [STUXNET] [HUA].

(3) Unavailability of hardware due to supply chain problems

Issues in the supply chain that lead to delay of hardware delivery may result in outage of MGMT system or at least performance loss of the system. This can be caused by pandemics, accidents or terrorism leading to disruptions in the global supply chain. Supplier monopoly may lead to bottlenecks in the supply chain and may cause vulnerabilities of the MGMT System to breach SLAs and therefore corresponding downtimes [ENISATHREAT] [SUEZ].

(4) Hardware wear can cause hardware failure and outages

System failures from aging hardware components may lead to unavailability of the MGMT System [ENISAINC] [E].

(5) Missing redundancy of hardware implementation

A complete lack of, or insufficient, redundancy concepts may cause key MGMT systems to be unavailable in case of hardware failure [ENISAINC] [E].

Probability Assessment

Due to the rarity of extreme natural disaster and the conditional probability of the MGMT system or any hardware being rendered unavailable by such an event, the overall probability for a MGMT system becoming unavailable due to failure of hardware components is evaluated as **LOW**. MGMT systems can be installed on any computing hardware which makes an unavailability of spare parts very unlikely. The most relevant attack path includes intentional manipulation or supply chain attacks on hardware. However, due to expected redundancy and the well protected server hardware, it is assumed that an attacker needs physical access to these systems to cause real hardware damage.

5.2 Public Mobile Network Operator (MNO) Perspective

This section discusses the risk scenarios related to the public MNO's perspective.

In each subsection, a short evaluation of the relevance of the risk scenarios in the MGMT is discussed and the probability that potential attack paths could be exploited is derived.

5.2.1 Emergency Call (TKG § 164)

The emergency call is a service that is an obligatory built-in function in public 5G networks. Any device that is attached to the AS of a network is allowed to perform emergency calls. Because it is an in-built base function, management systems that can configure the emergency call functions are either very limited in their amount or simply non-existent. Basically, there are no direct configurations necessary for the management regarding emergency calls. Consequently, the management system cannot directly affect essential services such as emergency calls, cell broadcasts or similar.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of misconfiguration in 5G management systems affecting the emergency call function is LOW . Even in a scenario where management systems become completely unavailable due to misconfiguration, the most severe outcome would be the inability to monitor and configure the 5G system. While unavailability of the MGMT system over extended periods will lead to accumulations of unresolved errors or security incidents which may finally cause service failure, a misconfigured MGMT system will not lead to an immediate failure of the emergency call functionality of the 5G system. Additionally, emergency services are deemed highly critical and are typically designed as robust and resilient systems to ensure their continuous operation. This is because it is required by law for public MNOs to “ensure or duly help to ensure that emergency calls can be made at any time and reach the correct local emergency service centre without delay” [BNETZA]. Following this reasoning, the probability of the risk is considered LOW .	Low
Insufficient Access Controls	Availability (AP2-A)	The risk probability of failure of emergency call due to insufficient access control within the public MNO perspective is LOW . Weak authentication, credential theft or identity abuse can give an attacker direct access to the MGMT system and creates a general risk for the system, not only affecting the availability of the system [SECENG, KYIV]. Privilege abuse and insecure APIs open the possibility to interact with the MGMT system in a malicious way disrupting the systems availability [SUPATK]. For a public MNO, this could hinder their ability to maintain reliable management systems. However, as the emergency call may not be impacted by an attacker gaining access to management systems, the probability of the risk due to insufficient access control is evaluated as LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of failure of emergency call due to implementation flaw within the public MNO perspective is LOW. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems. While unavailability of the MGMT system over extended periods will lead to accumulations of unresolved errors or security incidents which may finally cause service failure, a defective MGMT itself will not lead to an immediate failure of the emergency call functionality of the 5G system. Additionally, emergency services are deemed highly critical and are typically designed as robust and resilient systems to ensure their continuous operation. Following this reasoning, the probability of the risk is considered LOW.	Low
Specification Weakness	Availability (AP4-A)	The risk probability of failure of emergency call due to specification weakness within the public MNO perspective is LOW. The lack of official specifications for 5G management systems can lead to inconsistent implementation of security measures and architectural best practices by different vendors. This inconsistency may result in systems exhibiting varied behaviors, potentially compromising all security goals, including those related to emergency calls. Weaknesses in the specification can make attacks (e.g. DoS) on the MGMT system possible which could disrupt its functions. For a public MNO, this could hinder their ability to maintain reliable emergency call services. However, in general the MGMT system does not directly affect emergency call operations and the probability of such a risk is therefore evaluated as LOW.	Low
Failure of Power Supply	Availability (AP5-A)	The probability of a power supply failure in 5G management systems affecting the emergency call function is LOW. According to [ENISAINC], power supply failures or hardware failures are a significant risk in general. However, emergency services are deemed highly critical and are typically designed with robust, resilient systems to ensure their continuous operation, even if the management system is unavailable due to potential power supply issues. Unavailability of the MGMT system over extended periods will lead to accumulation of unresolved errors or security incidents, that might not be recognized and will increase the probability of emergency service failure. Anyway, an inaccessible MGMT system will not lead to an immediate or short time failure of the emergency call functionality of the 5G system, therefore, the probability of the risk is considered LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The risk probability of failure of emergency call due to failure of hardware components / unavailability of spare parts within the public MNO perspective is LOW. The risk of failure of emergency call in a 5G management system can arise from various hardware-related issues. For a public MNO, such failures can impede emergency call services, posing a serious risk to public safety. However, a failure of the MGMT hardware unlikely leads to a failure of emergency call service due to separation of the systems. Missing redundancy in hardware implementations could exacerbate the risk of emergency call failure. Failure of MGMT HW will not lead to immediate outage of the cellular system, but unavailability of the MGMT system over extended periods will lead to accumulation of unresolved errors or security incidents, that might not be recognized and will increase the probability of emergency service failure. This could lead to a failure of the system. Despite these potential issues, the probability of this risk is evaluated as LOW.	Low

5.2.2 Cell Broadcast (TKG § 164a)

Cell broadcast is an essential service in mobile networks, used primarily for delivering emergency alerts and public safety messages. It allows messages to be sent to all devices within a specific area without requiring active connections from the users. The service is highly reliable with minimal configuration, as it operates with predefined settings and is largely independent of broader network management systems. Therefore, the management system has limited influence over the functionality of cell broadcast, reducing the risk of service disruption. The probability assessment is identical compared to the risk assessment in section 5.2.1.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Cf. section 5.2.1 (Emergency Call)	Low
Insufficient Access Controls	Availability (AP2-A)		Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)		Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Availability (AP4-A)		Low
Failure of Power Supply	Availability (AP5-A)		Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)		Low

5.2.3 Telecommunications Secrecy (TTDSG § 3)

The management system itself, disregarding lawful interceptions that may or may not be implemented, does not and cannot directly access user data that is sent through the network. However, management systems store a lot of other data that is relevant to users' communication. Generally, examples of data stored in management systems are billing information, tariffs, general user information or user settings such as quality of service or subscribed tariffs or services options and finally user activity. On top of that, during provisioning process of new users, user identities such as the IMEI of the UE may be linked to the IMSI. While this does not allow direct access to the transmitted data of a user from the management system, such metadata of the user's communication can be leaked.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The risk probability of misconfiguration in 5G management systems impacting telecommunications secrecy is classified as MEDIUM. Due to the high likelihood of at least some errors in configuring endpoints or security algorithms in MGMT the probability of this risk is evaluated as MEDIUM. The complexity and largeness of public 5G networks and their sophisticated security protocols increase the risk of human errors, which can create vulnerabilities and weaken encryption. Thus, despite robust protocols, the inherent complexity and frequent updates in 5G technology and general IT-security necessary contribute to this probability estimation.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	The risk probability of failure of telecommunication secrecy due to insufficient access control within the public MNO perspective is MEDIUM. Insufficient access control in a 5G MGMT system pose a risk for telecommunication secrecy. An attacker exploiting weak authentication, stealing credentials, or escalating privileges can directly access sensitive user metadata such as call frequency, duration, and timing from the management system. This metadata has to be safed for billing and lawful interception purposes. Even though it does not contain plain user data, the sensitivity of such data is crucial. For a public MNO, unauthorized access to such data undermines trust and legal compliance, making it a critical risk source. The risk probability of telecommunication secrecy failure is especially high due to many known incidentts that proof that access to management systems is possible [SECSPEC, ATNT24, ATHENS, KYIV].	Medium
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The risk probability of implementation flaws in the 5G management system affecting telecommunications secrecy is classified as MEDIUM. This is because even unintentional implementation flaws may lead to accidental data leakage, as seen in cases like the Heartbleed vulnerability [HEARBL]. Implementation flaws can compromise security mechanisms or other aspects related to the protection of stored data, increasing the likelihood of unauthorized access and breaches. Additionally, intentional manipulation of software by e.g. supply chain attack [SOLAR] is a possible attack path. The latter are complex attacks, but the relatively high outcome makes it attractive for potential attackers. Public network operators are well aware of the sensitivity of meta data and have implemented counter measures accordingly. Because of those different angles the overall risk probability ist evaluated as MEDIUM.	Medium
Specification Weakness	Confidentiality (AP4-C)	The risk probability of failure of telecommunication secrecy due to specification weakness within the public MNO perspective is LOW. Specification weaknesses in protocols used by a MGMT system (such as TLS or IPSec) can lead to attackable protocols potentially leaking user related data [SECSPEC]. An exploitable specification of Lawful Interception (LI) can lead to unlawful interception of user communication [ATHENS]. However, sticking to state-of-the-art protocols and architectures, assuming up-to-date systems, the risk is considered as LOW.	Low
Failure of Power Supply	Confidentiality (AP5-C)	No relevant attack path could be identified.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	No relevant attack path could be identified.	Low

5.2.4 Emergency Preparedness (TKG § 184)

Emergency preparedness is a fundamental requirement in public 5G networks to ensure the availability of services during disasters. For instance, mobile network operators are legally obligated (§ 184 TKG) to prioritize predefined users in such events. As this is a critical function, management systems are often limited in their capability to apply configurations to systems relevant for emergency preparedness, as there is usually no configuration needed during an emergency. Such configurations are usually applied and tested beforehand of the system's commissioning. Consequently, a management system's availability has a limited influence on the prioritization mechanism.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of degraded emergency preparedness due to misconfiguration within the public MNO perspective is LOW. The full extent of an emergency is difficult to test in advance. However, operators are sensitive to such cases and are obliged to do so because of the legal basis (TKG). Furthermore, the configurations for emergency preparedness are defined and incorporated in advance so that there is no longer any dependency on management in the event of an emergency. Therefore, a misconfiguration of management system may delay or hinder the configuration of such emergency protocols. Because of the independence from the actual time of the emergency case, and the automation of these emergency procedures, omitting the need of manual input, the overall probability is LOW.	Low
Insufficient Access Controls	Availability (AP2-A)	The risk probability on emergency preparedness due to insufficient access control within the public MNO perspective is LOW. Weak authentication, credential theft or identity abuse can give an attacker direct access to the MGMT system. Privilege abuse and insecure APIs further open the possibility to interact with the MGMT system in a malicious way disrupting the systems availability [SUPATK]. However, as the emergency procedures may not be impacted by an attacker gaining access to management systems, the probability of the risk due to insufficient access control is evaluated as LOW.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The probability of an implementation flaw being the cause of a degraded emergency preparedness is LOW. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. Because the emergency preparedness is incorporated in advance, such flaws may hinder the configuration of emergency protocols, but won't prevent them over a longer time period. Nonetheless, emergency procedures are typically fully automated and don't need manual input via management systems. Therefore, the overall probability is considered as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Availability (AP4-A)	The risk probability on emergency preparedness due to specification weakness within the public MNO perspective is LOW . Weak or the absence of specification of 5G MGMT system can increase the risk of emergency preparedness by unclear interaction with the system by the operator or an undefined respond of the system. However, this issue is related to business specific implementations rather than 5G MGMT specific. Overall, the risk is considered LOW .	Low
Failure of Power Supply	Availability (AP5-A)	No relevant attack path could be identified.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	No relevant attack path could be identified.	Low

5.2.5 Regional Network Failure

While the management software controls and coordinates multiple network functions, disrupting or compromising the management system does not directly affect the underlying managed components or specifically a certain area of the mobile network. The 5G management system is mainly used for monitoring or configuring the 5G sub-components. The MGMT system does not need to actively control underlying components or certain areas of the mobile network immediately, therefore, in the short or medium term the underlying components are not relying on it. The MGMT system is only needed if configuration must be applied, or monitoring data needs to be accessed. Therefore, the case of a regional network failure is considered **LOW** for all attack paths. Also, the occurrence of a failure in a smaller part of the network, e.g. in a single gNB, is higher than the simultaneous failure of multiple gNBs distributed across multiple regions. Since the risk for multi-regional network failure is also considered **LOW**, in practice the risk for a regional network failure is considered to be even lower.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of misconfiguration causing regional network failure in the 5G system is LOW. Only misconfiguration of the management system itself is considered. Even in the high-impact scenario where the entire management system becomes unavailable, preventing new configurations or monitoring of the underlying network, the immediate functionality of the underlying network still remains largely unaffected. Only through a longer outage of the MGMT and accumulation of errors in the network the inaccessibility of the MGMT system will eventually lead to a regional network failure. This time window provides an opportunity to address the issue before it affects the broader network. Therefore, the probability is considered LOW.	Low
Insufficient Access Controls	Availability (AP2-A)	The risk probability of regional network failures due to insufficient access control within the public MNO perspective is LOW. Weak authentication, credential theft or identity abuse can give an attacker direct access to the MGMT system and create a general risk for the system including its availability [SECENG]. Privilege abuse and insecure APIs open the possibility to interact with the MGMT system in a malicious way disrupting the systems availability [SUPATK]. For a public MNO, this could mean a risk of regional network failures. However, because we do not consider the access to the 5G components and sub-systems over the management system but only the access to the management system itself, the probability of regional network failures due to this is considered as LOW.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of implementation flaws causing regional network failure in the 5G system is LOW. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems. Because of the separation of management and execution, an outage of the management system won't directly impact the availability of a regional network. Therefore, the probability is considered LOW.	Low
Specification Weakness	Availability (AP4-A)	The probability of the risk that weaknesses in the specifications will cause a regional network failure is classified as LOW. Specifications for the management system are almost non-existent, therefore, MGMT systems might vary in their architecture and behaviour. However, it is very unlikely that serious conflicts will arise due to a missing or weak specification. Availability is an important objective for telecommunications networks, to which operators pay particular attention. Hence, they design their management system implementation to promote system availability. As there is no direct impact from the specification of the MGMT system to the regional network the probability of the risk is evaluated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The risk of failure of power supply being the root cause of regional network failure is categorized as LOW. Failure of power supply may be caused by natural disasters, accidents or terrorist attacks. In this analysis we only consider power outages of the management system, but not the power supply failure of the remaining 5G system. Failure of power supply might render the management system unavailable, preventing an operator from applying new configurations. However, it might take a relatively long time before the inaccessibility of the MGMT system leads to regional network problems. This time window provides an opportunity to address the issue before it affects the broader network. Also, the management system of public MNOs typically runs in resilient datacenters where also the other mobile services are hosted, therefore a failure of power supply is very unlikely [BSIRZHV] and the probability of this scenario is considered as LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The risk probability of failure of regional networks due to failure of management hardware components / unavailability of spare parts within the public MNO perspective is LOW. The risk of failure of regional networks in a 5G management system can arise from various hardware-related issues. Failure of MGMT HW will not lead to immediate outage of the cellular system, but unavailability of the MGMT system over extended periods will lead to accumulation of unresolved errors or security incidents, that might not be recognized and therefore increase the probability of a regional network failure. Despite these potential issues, the probability of this risk is evaluated as LOW.	Low

5.2.6 Multi-Regional Network Failure

Just as in the case of regional network failure, any disruption or compromise to the management system itself still does not directly affect the underlying managed components or network, although the management software controls and coordinates multiple network functions. The 5G management system is mainly used to monitor and configure underlying 5G sub-components. The MGMT system does not need to actively control underlying components or regional networks immediately, therefore, in the short or medium term the underlying components are not relying on it. The difference to the Regional Network Failure scenario is here, that for the impact to be multi-regional multiple regional failures or more severe failures have to occur. Therefore, the risk for a failure of the mobile network in a larger geographical area is in practice lower than for a regional network failure, although the probability level in this analysis is the same, because of the coarse granularity.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of misconfiguration causing multi-regional network failure in the 5G system is LOW. Only misconfiguration of the management system itself is considered. Even in the high-impact scenario where the entire management system becomes unavailable, preventing new configurations or monitoring of the underlying network, the immediate functionality of the underlying network still remains largely unaffected. Only through a longer outage of the MGMT and accumulation of errors in the network the inaccessibility of the MGMT system will eventually lead to a multi-regional network failure. This time window provides an opportunity to address the issue before it affects the broader network. In general, the probability is just marginal higher than in the regional network failure. Assuming that redundancy and systems are in place to quickly regain availability in the case of MGMT outages the risk probability is considered LOW.	Low
Insufficient Access Controls	Availability (AP2-A)	The risk probability of multi-regional network failures due to insufficient access control within the public MNO perspective is LOW. Weak authentication, credential theft or identity abuse can give an attacker direct access to the MGMT system and create a general risk for the system including its availability [SECENG]. Privilege abuse and insecure APIs open the possibility to interact with the MGMT system in a malicious way disrupting the systems availability [SUPATK]. For a public MNO, this could mean a risk of multi-regional network failures. However, because we do not consider the access to the 5G components and sub-systems over the management system but only the access to the management system itself, the probability of multi-regional network failures due to this is considered as LOW.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of implementation flaws causing multi-regional network failure in the 5G system is LOW. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems. Because of the separation of management and execution, an outage of the management system won't directly impact the availability of larger areas of the mobile network. Therefore, the probability is considered LOW.	Low
Specification Weakness	Availability (AP4-A)	The probability of the risk that weaknesses in the specifications will cause a multi-regional network failure is classified as LOW. The specifications for the management system are almost non-existent, therefore, it is very unlikely that serious conflicts will arise. Availability is an important objective for telecommunications networks, to which operators pay particular attention. Hence, they design their management system implementation to promote system availability. As there is no direct impact from the specification of the MGMT system to the regional network the probability of the risk is evaluated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The risk probability of failure of power supply being the root cause of multi-regional network failure is categorized as LOW. Failure of power supply may be caused by natural disasters, accidents or terrorist attacks. In this analysis we only consider power outages of the management system, but not the power supply failure of the remaining 5G system. A failure of power supply may render the management system unavailable, preventing new configurations. For multi-regional network failures to occur, there must be multiple simultaneous errors or more severe errors than for the failure of regional Networks. Additionally, it again might take a relatively long time before the inaccessibility of the MGMT system will lead to such widespread issues, making the probability even lower.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The risk probability of failure of multi-regional networks due to failure of management hardware components / unavailability of spare parts within the public MNO perspective is LOW. The risk of failure of multi-regional networks in a 5G management system can arise from various hardware-related issues. Failure of MGMT HW won't lead to immediate outage of the cellular system, but unavailability of the MGMT system over extended periods will lead to accumulation of unresolved errors or security incidents, that might not be recognized. This could lead to a failure of the system. Despite these potential issues, the probability of this risk is evaluated as LOW.	Low

5.2.7 Failure of Mobile Service

Mobile services, like voice calls, data transmission or edge applications, are operated by the 5G Core and are highly dependent on the availability of the core system itself. In cellular networks such services are not generally available to all network users and are therefore interconnected to a user access rights system, that lies within the management domain. Albeit tampering with the management system may not interfere with the stability of the service providing system itself, but with the accessibility to certain services. This makes the risk scenario of mobile service failures highly relevant for the management domain.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of misconfiguration causing failures in mobile services in the 5G system is MEDIUM. This is because there are many services, some of which are related to the management system or constantly need information on different users' access permission from the management system. If the management system fails due to misconfiguration, it can cause disruptions in these interconnected services. The complexity and interdependence of mobile services increase the likelihood of issues arising from such misconfigurations, hence the medium probability. For example, we could consider normal phone calls. During the call setup procedure the core network must get information on e.g. voice service usability or billing information from the management system. If the management system is not available in this situation or misconfigured, the mobile service might not be available for a specific user or for groups of users. According to [ENISAINC] and [E] the probability of human errors or malicious actions regarding mobile services is medium.	Medium
Insufficient Access Controls	Availability (AP2-A)	The risk probability of failure of mobile service due to insufficient access control within the public MNO perspective is MEDIUM. Insufficient authentication in 5G management systems can lead to various attack paths, including exploitation of weak authentication, credential theft, identity abuse, and privilege escalation. For an MNO, this increases the risk of service failures, as attackers could manipulate tariffs making the service unavailable to users. The root cause of this risk is the lack of robust authentication mechanisms. The probability of such service failures occurring due to insufficient access control is evaluated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of implementation flaws causing failures in mobile services in the 5G system is MEDIUM. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems, as seen during CrowdStrike incident [CROWDS]. Because mobile services may need to obtain user permissions, that are managed by the BSS part of the MGMT System, nonreachability of this BSS may lead to unavailability of such services. Since the risk for availability problems due to implementation flaws is relatively high in general, the risk for an impact on the MGMT System leading to problems for the mobile services is considered MEDIUM.	Medium
Specification Weakness	Availability (AP4-A)	The risk probability of failure of mobile service due to specification weakness within the public MNO perspective is LOW. Weak or the absence of specification of 5G MGMT system can increase the risk of failure of mobile service by unclear interaction with the system by the operator or an undefined respond of the system. However, this issue is related to business specific implementations rather than 5G MGMT specific. Overall, the risk is considered LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The risk probability of failure of power supply causing failures in mobile services in the 5G system is LOW. Although the situation of failure of power supply might happen quite often in mobile networks regarding to [ENISAINC], in the case of the management system the probability is considered low. This is because the management system of public MNOs typically runs in highly resilient datacenters where also the other core mobile services are hosted, therefore a failure of power supply is very unlikely [BSIRZHV]. Most likely, outages will appear on a smaller regional scale not in the core datacenters. Consequently, the probability of failure of mobile services due to failure of power supply of the MGMT system is considered LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The risk probability of failure of hardware causing failures in mobile services in the 5G system is LOW. The management system needs to be available in order to adjust billing information and service access rights for users. If the management system is unavailable, no changes in tariff and service can be done. While user rights management for services is part of the management domain, a long- or mid-term unavailability of the management caused by hardware failures won't directly impact the current user rights configuration already established in the 5G system. Failure of the management system is also unlikely due to typical redundancy measures. Being deployed on standard COTS hardware and virtualization solutions provides the ability of migrating virtualized environments, further decreasing the probability of failure of mobile services due to failure of hardware components of the MGMT system to LOW.	Low

5.2.8 Fraud

This risk scenario assumes that an attacker manipulates user tariff or billing data to gain access to mobile services in order to gain monetary benefits. Because the management system handles user rights and billing information through the BSS this scenario is highly relevant for the management system. Still this risk scenario assumes a small-scale operation, since it also needs to remain unnoticed by operators.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The probability of fraud due to misconfiguration of the 5G MGMT system is MEDIUM. Unintentional manipulation can be ruled out by the definition of fraud. If the ability to configure the management system and especially the BSS is given, misconfigurations, e.g., changes to subscriber contracts can easily be made. But because of the sensitivity of such data, public network operators have a high incentive in protecting their customer contracts from external attacks, decreasing the probability of malicious access. Internal attackers may still use their access privileges in order to gain financial benefits [INTERNAL]. The mitigation strategies for those internal attacks might not be that enhanced, but on the other hand the punishment for abusing access rights might be high. So especially for insiders it will be in theory easily possible to commit fraud. Therefore, the overall risk probability is evaluated as MEDIUM.	Medium
Insufficient Access Controls	Integrity (AP2-I)	The risk probability of fraud due to insufficient access control within the public MNO perspective is MEDIUM. Insufficient authentication in 5G management systems can lead to various attack paths, including exploitation of weak authentication, credential theft, identity abuse, and privilege escalation. For a public MNO, this increases the risk of fraud, as attackers could manipulate billing systems and violate the integrity of them very easily by exploiting possible insufficient access control if no robust authentication mechanisms are in place. The critical nature of insufficient access control and the direct possibility to act fraudulent makes such an attack likely, however, access to MGMT systems is not easily gained. Therefore, the overall risk probability is evaluated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	The probability of fraud due to implementation flaws of the 5G system is LOW. In most cases implementation flaws are unintentional and lead to system errors or data leakage. Enabling fraudulent behavior by this kind of flaw is highly unlikely. In order to commit fraud, intentional manipulation of software by e.g. supply chain attack [SOLAR] is necessary. The high complexity and resource requirements make this kind of attack very difficult to carry out. The overall risk probability is therefore evaluated as LOW.	Low
Specification Weakness	Integrity (AP4-I)	The risk probability of fraud due to specification weakness within the public MNO perspective is LOW. In the 5G management system, specification weaknesses, such as missing or insufficient protocol definitions between management components, can impact fraud risk. For a public MNO, these vulnerabilities can lead to unauthorized access and manipulation of billing information, compromising the integrity of the billing service. While the probability of this risk due to specification weakness is low due to the typically robust security measures in place, the root cause is the inadequate specification of communication protocols. This can result in undetected fraudulent activities and financial losses [SOKFRAUD, GTPDOOR]. Ensuring precise and comprehensive protocol specifications is crucial to mitigate this risk and protect the billing service's integrity.	Low
Failure of Power Supply	Integrity (AP5-I)	No relevant attack path could be identified.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	No relevant attack path could be identified.	Low

5.2.9 Subscriber Data Leak

In the management system, a lot of privacy sensitive data of its connected users and UEs is stored or can be accessed. Examples of such privacy sensitive data are billing, tariffs, general user information or user settings such as quality of service or subscribed services. On top of that during provisioning of SIM-Cards, the user identity can be linked to the IMSI/SUPI. In contrast to other risk scenarios such as telecommunication secrecy this case does not include the risk probability of data transmitted by the user. Because of the sensitivity of subscriber data, and the high number of associated fines, public MNOs have a high incentive to protect this kind of data. Therefore, security measures, such as encryption, firewalls, virtual private networks or employee training should be in place to protect the system from leaks, as required by the General Data Protection Regulation (GDPR), which was enacted by the European Union in 2018 [GDPREU]. However, at the same time this data is very attractive for attackers as well.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The probability of misconfiguration in 5G management systems causing subscriber data leak is classified as MEDIUM. This is due to the high likelihood of errors in configuring endpoints [TELEAPI] [TELEAPI2] or security algorithms in the 5G management system. The complexity of 5G management and sophisticated security protocols increase the risk of human errors, which can create vulnerabilities and weaken encryption. Thus, despite robust protocols, the inherent complexity and frequent updates in 5G technology and general state of IT-security necessary contribute to this probability estimation.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	The risk probability of subscriber data leakage due to insufficient access control within the public MNO perspective is MEDIUM. Insufficient access control in a 5G MGMT system pose a significant risk for subscriber data leaks. Weak protocols, insufficient physical or remote access controls, and poor authentication practices can grant attackers access to sensitive data, leading to potential subscriber data leaks. This includes exploitation of weak passwords, absence of multi-factor authentication, use of default credentials, and insecure handling of public key infrastructure. Credential theft and identity abuse through phishing or keyloggers, injection of malicious software or hardware, insecure APIs, and privilege escalation further expose the management system to unauthorized access. Once access is gained, the attacker can simply view the subscriber data of every subscriber. The risk probability of subscriber data leaks is especially high due to many known incidents that prove that access to management systems is possible and the high benefit for attackers in terms of ransom attacks or others [SECSPEC, ATNT24, ATHENS, KYIV, TELEAPI1]. Public network operators are well aware of the sensitivity of subscriber data and have implemented counter measures accordingly. Because of that, the overall risk probability is evaluated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The risk probability of implementation flaws in the 5G management system leading to subscriber data leak is classified as MEDIUM. This is because even unintentional implementation flaws may lead to accidental leakage of user data, as seen in cases like the Heartbleed vulnerability [HEARBL] and [TELEAPI]. Implementation flaws can compromise security mechanisms or other aspects related to the protection of stored data, increasing the likelihood of unauthorized access and breaches. Additionally, intentional manipulation of software by e.g. supply chain attack [SOLAR] is a possible attack path. The latter are complex attacks, with high resource requirements, therefore a very strong attacker is needed to carry them out. Public network operators are well aware of the sensitivity of subscriber data and have implemented counter measures accordingly. Because of that, the overall risk probability is evaluated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Confidentiality (AP4-C)	The risk probability of subscriber data leakage due to specification weakness within the public MNO perspective is LOW. The risk of "Subscriber Data Leakage" in a 5G Management System can also arise due to insufficient specification. Without official specifications, vendors might implement varying security measures, causing inconsistencies that affect security goals. Specifically, the lack of adequate specifications for authentication and encryption protocols between management components can result in data being observed, compromising confidentiality of subscriber data. Moreover, weaknesses in protocol specifications such as TLS or HTTP, and exploitable specifications in Lawful Interception systems, further increase the risk. For a public MNO, these specification weaknesses could lead to unauthorized access and leakage of subscriber data. Overall, the probability of such an attack path is evaluated as LOW as manufactures usually provide state-of-the art implementations mitigating the probability of the risk.	Low
Failure of Power Supply	Confidentiality (AP5-C)	No relevant attack path could be identified.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	No relevant attack path could be identified.	Low

5.3 User Perspective

This section describes different risk scenarios from the perspective of a user of the public mobile network. Private 5G networks are not in scope of this section. Since some of the typical risk scenarios (e.g. emergency calls) have already been described in the previous paragraphs, they won't be mentioned again in this section. In general, the management system is a very sensitive asset for mobile network operators. As such, both, the general security as well as the legal consequences for attackers is very high.

5.3.1 Urgent Telephone Calls Not Possible

This describes the simple risk that an urgent call cannot be made due to unavailability of the public mobile network.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of misconfiguration causing urgent telephone calls to not being possible is LOW. The probability of the related risk scenario "failure of mobile services" was classified as medium. In the case of urgent telephone calls not being possible, only one particular service, the voice service, is considered. Therefore, the probability of this exact service failing or being unavailable due to misconfiguration must in consequence be lower or equal than failure of any mobile service. Since the voice service is the oldest and most tested mobile service, many of the possible problems leading to unavailability of the service have already been unveiled once and countermeasures have been implemented. Also, misconfiguration of the management system can only have minimal impact on the availability of the mobile telephone service, which is mostly completely automated. Possible disruptions of urgent telephone calls originate from a long-term unavailability of management system or from misconfiguration of the customer contracts, e.g., in order to revoke rights to use voice services. Even though those cases are possible, they are very unlikely to happen by accident.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	From a user perspective, the probability of urgent telephone calls not being possible due to insufficient access control is LOW. The risk of unavailability of general mobile services is medium and therefore the probability for the telephone call service failing is equal to that at most. Even though, there are a couple of attack paths, as weak authentication, credential theft or identity abuse can give an attacker direct access to the MGMT system and create a general risk for the system, not only affecting the availability of the system [SECENG, KYIV]. Privilege abuse and insecure APIs open the possibility to interact with the MGMT system in a malicious way disrupting the systems availability [SUPATK]. For a public network user, this could hinder their ability to maintain reliable call services. However, as telephone calls are not directly impacted, because the corresponding functions are not part of the MGMT, but part of the core network, the probability of such a risk is evaluated as LOW.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of implementation flaws causing urgent telephone calls to not be possible is LOW. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems. While unavailability of the MGMT system over extended periods will lead to accumulations of unresolved errors or security incidents which may finally cause service failure, a misconfigured MGMT in itself will not lead to an immediate failure of the voice call functionality of the 5G system. In addition, intentional manipulation of software by e.g. supply chain attack [SOLAR] is a possible attack path. But the high complexity and resource requirements make this kind of attack very difficult. Following this reasoning, the probability of the risk is considered LOW.	Low
Specification Weakness	Availability (AP4-A)	The risk probability of urgent telephone calls not being possible due to specification weakness within the user perspective is LOW. The lack of official specifications for 5G management systems can lead to inconsistent implementation of security measures and architectural best practices by different vendors. This inconsistency may result in systems exhibiting varied behaviors, potentially compromising all security goals, including those related to telephone calls. Although the MGMT system does not directly affect voice call operations, weaknesses in the specification can make attacks (e.g. DoS) on the MGMT system possible which could disrupt its functions, indirectly impacting emergency call handling [ESEC, GTPDOOR]. For a public network user, this could hinder their ability to maintain reliable voice services. However, as urgent telephone calls are not directly impacted the probability of such a risk is evaluated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The risk probability of failure of power supply causing urgent telephone calls to not be possible is LOW. The probability of urgent telephone calls not being possible due to a power supply failure of the management system is classified as LOW. The mobile telephone service is not directly affected by the management system, therefore, an unavailable management system does not hinder urgent telephone calls. Additionally, the IT infrastructure of public MNOs is supposed to run in highly redundant, well protected datacenters [BSIRZHV]. If one server fails due to a failure of power supply another server will probably take over the task instantly or in a short period of time. Such redundancy and failover mechanisms are best practices and are usually implemented in any data center. Since those servers are distributed to different locations the probability of multiple power supply failures affecting all of the servers at once is very low. Therefore, the overall probability of failure of power supply leading to unavailability of the MGMT system leading to a situation where important calls cannot be made is very unlikely and the risk probability is considered LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The risk probability of urgent telephone calls not being possible due to failure of hardware components / unavailability of spare parts within the user perspective is LOW. The risk of failure of telephone calls in a 5G management system can arise from various hardware-related issues. For a public network user, such unavailability can impede voice services. However, it is unlikely that a failure of the MGMT hardware and its resulting unavailability leads to an immediate failure of call services due to separation of the systems and the missing impact from the MGMT system on the IMS. Missing redundancy in hardware implementations could exacerbate the risk of voice call failures. Failure of MGMT HW won't lead to immediate outage of the cellular system, but unavailability of the MGMT system over extended periods will lead to accumulation of unresolved errors or security incidents, that might not be recognized. This could lead to a failure of the system. Despite these potential issues, the probability of this risk is evaluated as LOW.	Low

5.3.2 Authenticity of the Receiver of a Call

Changing the identity of call receivers is mainly used in the so called “call forwarding scam”. The standard attack path for this scam is to manipulate the user and change the call forwarding settings from the UE side. Another, more complex and harder to accomplish attack path, may be achieved by manipulation of call forwarding settings directly within the Home Location Register. The latter case will be considered here.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The probability of losing authenticity of the receiver of a call due to misconfiguration of the management system is LOW. A relevant misconfiguration, whether malicious or not, would for instance be the manipulation of call forwarding configuration that is changed within the Home Location Register (HLR). For a public network user, this compromises the authenticity of the receiver of a call, enabling call forwarding scams [CALLFOR]. The impact on call authenticity is significant, making it difficult to trust the identity of call recipients. Although this attack path might be possible in principle, the probability of this risk occurring is LOW due to the high effort and complexity involved in creating such scams. Also, the probability of accidental misconfiguration leading to problems for the authenticity of the receiver of a call is very low.	Low
Insufficient Access Controls	Integrity (AP2-I)	The risk probability of losing authenticity of the receiver of a call due to insufficient access control within the user perspective is LOW. Several attack paths, including weak authentication, credential theft, insecure APIs, and privilege escalation may lead to insufficient access control. These vulnerabilities can allow attackers to access sensitive user data in the underlying core system indirectly through the management domain. For instance, an attacker could access user data containing call forwarding information that is stored within the Home Location Register (HLR). For a public network user, this compromises the authenticity of the receiver of a call, enabling call forwarding scams [CALLFOR]. The root cause is inadequate access control mechanisms. The impact on call authenticity is significant, making it difficult to trust the identity of call recipients. As an attacker first needs to gain access to the management system and then execute complex configurations, which is a high effort, the overall probability of this risk occurring is LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	The probability of losing authenticity of the receiver of a call due to implementation flaws in the management system is LOW. In most cases implementation flaws are unintentional and lead to system errors or data leakage. Enabling manipulation of call forwarding information by this kind of flaw is highly unlikely. In order to perform call forwarding scams, intentional manipulation of software by e.g. supply chain attack [SOLAR] is necessary. But the high complexity and resource requirements make this kind of attack very difficult to carry out. The overall risk probability is therefore evaluated as LOW.	Low
Specification Weakness	Integrity (AP4-I)	The risk probability of losing authenticity of the receiver of a call due to specification weaknesses within the user perspective is LOW. Specification weaknesses, such as insufficient management system specifications, inadequate protocol definitions between management components, and exploitable lawful interception (LI) protocols, can lead to significant vulnerabilities. For a public network user, this results in uncertainty, if more than one call recipient exists, as attackers might exploit these weaknesses to intercept calls [ATHENS]. The root cause is poorly defined specifications and protocols. The probability of this risk occurring is LOW due to the complexity and legal obligations to use these systems and the effort that is usually put into designing such protocols, in a best case, having them formally verified.	Low
Failure of Power Supply	Integrity (AP5-I)	No relevant attack path could be identified.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	No relevant attack path could be identified.	Low

5.3.3 Integrity of Transmitted Data

User data is not transmitted through the MGMT system. The **MGMT** is therefore **not considered** as a relevant risk source for integrity of transmitted data.

5.3.4 User's Identity

This risk scenario is about an attacker being able to steal the identity of different users and use these identities for example for purchases or other illegal activities. User identity in current 5G systems is mostly authenticated during a procedure called Authentication and Key Agreement (AKA). While there might be alternative authentication protocols, currently 5G authentication is based on the 5G AKA and EAP-AKA', which use mutual authentication through two shared permanent keys and UEs identity. The SUPI is hereby the identifier of the UE, just as the IMSI in former generations, that is used to initiate the authentication process with the 5G home network. This SUPI can simply be viewed by anyone that has access to the MGMT system. The permanent keys, shared by the UE and the home network, on the other hand cannot simply be accessed in the MGMT system.

To masquerade or spoof a user identity in a 5G network the two permanent keys would need to be eavesdropped. They rest in the physical SIM of the UE and the 5G Core function UDM (or HSS for 4G Core systems). The management system generally has no direct access to either of these. However, depending on the management system of the core, the provisioning of the keys may be done through the management system. If this process is eavesdropped this can potentially lead to a spoofing of user identities. Furthermore, since the exact functions of the MGMT systems and how they are implemented are not publicly available, it is uncertain if there exists a MGMT that can access the keys after provisioning. Generally, this is not the case though, and to be clear it is not possible to read key information after provisioning and it will therefore not be considered here [E].

This only leaves three relevant attack paths, which are highly sophisticated and/or need complete access of the MGMT. The first attack path simply includes a DoS attack on any UE by tampering with the users stored in the core. The second includes malicious provisioning, by adding a new user to the network which enables an attacker to add malicious devices to the network. These two attack paths are not relevant for the evaluation of this risk scenario because it is not about DoS or infiltration but precisely about spoofing a user identity. Lastly, the third attack path can be considered relevant to the observed risk scenario and is the one that is evaluated here. In this case the permanent shared keys are eavesdropped during the provisioning and then used in conjunction with a new SIM card and a malicious UE. It is to note that this case is very difficult and sophisticated.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The risk probability of misconfiguration of the 5G management system affecting integrity of user's identities is LOW. As stated in the preface, the management itself does not allow access to any key material or certificates necessary to steal a user's identity. Furthermore, the case of causing unavailability or adding malicious devices to a network is not covered by this risk scenario. Consequently, misconfigurations of the management itself cannot enable spoofing attacks on the user identity. But since only the configuration of the management system itself is considered here we categorize the risk for a breach of the integrity of a user's identity as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Integrity (AP2-I)	The risk probability of insufficient access control of the 5G management system affecting integrity of user's identities is LOW. As stated in the preface the management itself does not allow access to any key material or certificates necessary to steal a user's identity. Furthermore, the case of causing unavailability or adding malicious devices to a network is not covered by this risk scenario. Although what must be considered is the stealing of shared keys through the core management system by eavesdropping on the provisioning process as explained in the preface. However, this attack is very unlikely due to the complexity and many steps involved in such an attack. Therefore, the overall probability is considered LOW.	Low
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	The risk probability of implementation flaws in the 5G management system affecting integrity of user's identities is LOW. User Identities are only accessible by the management system during the small timeframe of the provisioning process. Because most flaws are unintentional and lead to system errors or unpredictable data leakage, an opportunity of user spoofing stemming from this fact is highly unlikely. Instead intentional manipulation of software by e.g. supply chain attack [SOLAR] in order to gain access to the MGMT during provisioning is necessary. Since such attacks have high complexity and resource requirements they are very difficult to be carried out. Overall this leads to the risk probability being considered as LOW.	Low
Specification Weakness	Integrity (AP4-I)	The risk probability of leak of user's identity due to specification weakness within the user's perspective is LOW. Specification weaknesses in protocols used by a MGMT system (such as TLS) can lead to attackable protocols potentially leaking user related data such as the user identity [SECSPEC]. As described in the preface, there is no way of tampering with the integrity of users' identities through the management system itself, therefore, weaknesses in third party protocols cannot cause direct impact on the user's identity. However, as there is no specification available, the assumption from the preface might not hold in a real scenario, heavily depending on the actual implementation of the operator. Management systems might exist which store all key material and certificates of users in its database, thus, making management systems vulnerable for these attacks on users' identities. If this is the case the absence of any specification is a major weakness because it "allows" operators to forego best practices of cryptographic key management. The overall risk probability is still considered to be LOW because MNOs typically follow best practices and regularly perform audits.	Low
Failure of Power Supply	Integrity (AP5-I)	No relevant attack path could be identified.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	No relevant attack path could be identified.	Low

5.3.5 Eavesdropping of a Communication

According to the description, the risk scenario for an eavesdropping attack on communication links relates to the connectivity between two participants. In this case, the confidentiality of the connection is not guaranteed. Confidential content may be accessible by a third party.

In contrast to other elements such as the RAN (data transmission via the air interface) or the core (management and forwarding of communication data; backbone for the confidentiality of data transmission via encryption and authorization mechanisms that protect communication from eavesdropping), a management system of a 5G network is not considered as relevant for processing the user data plane of any communication link by design. Then, vulnerabilities or shortcomings that allow access to the management system could not be exploited to gain unauthorized access to private conversations and breach confidentiality.

In fact, management systems are designed to ensure a clear separation of management functionalities from core functionalities like handling individual connections, see Figure 1. This is motivated by the fact that it is legally required to protect the user data, and MNOs are extremely sensitive in this regard when it comes to protecting this data [GDPREU]. Data leakage, whether intentional or unintentional, destroys a company's reputation and means not only an immense loss of image, but also considerable financial losses and years of legal disputes, potentially ruining the company. To avoid this catastrophic scenario, MNOs put a lot of effort into securing their systems.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The risk of misconfiguration leading to the possibility to eavesdrop on a communication is LOW. As described in the table head there is no communication routed through the management system and there is a strict separation between the management system and systems that handle user communication, there are only a few hypothetical ways misconfiguration of the management could possibly impact the confidentiality of communications. Depending on the implementation the lawful interception functionalities might be able to interact with the communication content. Since those services are very critical, they are also protected with high effort. Even if there is a way to interact with the content of user communication an attack path being able to utilize this is very unlikely to exist.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	The probability of the risk of being able to eavesdrop on communication links via insufficient access controls of the management system is classified as LOW. Insufficient physical or remote access controls can empower internal and external attackers to have access to the 5G management system and thus gain access to meta information of users [INTERNAL]. However, management systems do not provide direct access to the content of an active communication link due to strict separation of the systems. Therefore, to make eavesdropping via the MGMT system possible, the separation of management and core functionalities needs to be removed. Potential attacks are injections of malicious software/hardware during any process in the supply chain that may open backdoors to enable eavesdropping [HUA] [EDSNOW] [HTROJ]. Combined with insufficiently secured APIs [SUPATK], management functionalities could be exploited for eavesdropping attacks. Certainly, management systems could be set up as eavesdropping systems by design. Nevertheless, in order to be able to use a management system with inadequate access controls for eavesdropping [NSA], many preconditions are required, which are considered extremely unlikely based on the motivation of MNOs to secure user data, the security processes at MNOs in place and the constant development of these processes to remain secure.	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The risk of implementation flaws leading to the possibility to eavesdrop on a communication is LOW. As there is no direct connection to the communication channels from the management system eavesdropping of a communication is very unlikely. Therefore, implementation flaws only have a minimal chance of being able to allow an attack path that influences the confidentiality of user communication.	Low
Specification Weakness	Confidentiality (AP4-C)	The probability of the risk of being able to perform an eavesdropping attack on communication links caused by weaknesses in the specification is classified as LOW. In the context of 5G Management no official specification exists. Therefore, it is not guaranteed, that certain security measures or architectural best-practices are implemented in an arbitrary system being provided by different vendors. Furthermore, protocols like TLS or HTTP may contain weaknesses due to their specification that might lead to issues for all three security goals [GTPDOOR]. Finally, lawful interception (LI) systems might be an entry point for other attacks [ATHEN], where weaknesses in their specification could lead to exploitations and, hence, unallowed usage and observation of UEs including eavesdropping. Nevertheless, MNOs usually put a lot of work into the design of a system in order to meet safety targets. They are extremely critical of unknown systems and put them through rigorous testing before putting them into operation. Corresponding state-of-the-art solutions involve a clear separation of management and connectivity functionalities. Hence, a management system should not even provide direct access to the content of active communication links.	Low
Failure of Power Supply	Confidentiality (AP5-C)	No relevant attack path could be identified.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	No relevant attack path could be identified.	Low

5.3.6 Slowing Down/Restricting the Connection

A slow or restricted connection for an UE can certainly be caused by an issue with the 5G management system. Tariff and user service options related to the quality of service (including bandwidth, overall amount of data traffic as well as priorities) can be accessed from the management system making it a relevant entry point for the mentioned risk scenario.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of degraded services or connections of a single user due to misconfiguration in the management system is MEDIUM. The general service and prioritization of network traffic can be influenced over the management system through parameters of service classes and user groups, or on a larger scale, tariff settings. These can cause a downstream effect that change or influence core, RAN and finally user settings which result in slowing down of services. The risk probability for this attack path is categorized as MEDIUM. This is because a change to the settings can happen if contracts get initialized or changed. But it is in exactly those situations that users and providers also pay close attention to ensure the service is working as intended. Therefore, even though misconfigurations sometimes happen it is also likely for them to be detected. Other types of misconfiguration can cause delays or unavailability in the update of relevant parameters, for example endpoint misconfigurations or energy saving settings. Furthermore, misconfigurations of the security settings can be exploited by attackers.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	The probability of degraded services or connections of a single user due to insufficient access control in the management system is MEDIUM. Initial access to the necessary management systems could most likely be gained due to weak authentication abuse [SECSPEC] [ATNT24] [Kyiv], privilege escalation or simply insufficient access control and credential theft [ENISA], for example an insider using an unlocked laptop of a colleague to update his own service settings. Another attack could be injection attacks due to insecure APIs [SEEAPI]. Once access is gained, most likely attackers would influence user settings to cause a service degradation. After initial access is gained to the system the attack can be executed with relative ease. While it is difficult for attackers to gain access to management systems, the attack probability is considered as MEDIUM due to the ease of executing the attack once access is gained.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The probability of degraded services or connections of a single user due to implementation flaws in the management system is LOW. A disruption of an attacker that affects the management systems availability has in the short and medium time span no effect on the quality of service of the underlying communication of the mobile network. Consequently, only implementation flaws that affect the integrity of overlying parameters that steer the quality of the communication itself are relevant. Next to zero-day exploits that allow the tampering of data over secure protocols, especially outdated external software that is used for management software systems may enable attackers to tamper with subscriber data for quality-of-service assurance. But since the management system of public MNOs is a crucial component, outdated software is expected to be updated very fast by professional engineers, leaving only a minimal attack window open. Generally, attackers that use zero-days use it for larger scale of attacks. Therefore, the overall risk probability is considered LOW.	Low
Specification Weakness	Availability (AP4-A)	The probability of degraded services or connections of a single user due to specification weakness in the management system is LOW. The lack of official specifications for 5G management systems can lead to inconsistent implementation of security measures and architectural best practices by different vendors. This inconsistency may result in systems exhibiting varied behaviors, potentially compromising all security goals, including those related to service and connection handling. While weaknesses might be extrapolated because of missing specifications that can cause DoS or insufficient access control it does not directly influence the quality of service of the communication of the underlying network resulting in a LOW probability.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The probability of degraded services or connections of a single user due to failure of power supply in the management system is LOW. A failure in the power supply of the management systems means that access only to these specific systems is no longer possible. Changes to the underlying subscription data e.g. through an interrupted write process is highly unlikely. The management system itself has in a short and medium time no direct influence on the functionality of the core and RAN. Therefore, influence on the quality of the communication is not expected for short timeframes. But in a longer time period through the accumulation of errors, degradation and finally failure of communication services become more likely. Since such long time power outages are very unlikely, the overall risk probability is considered as LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of degraded services or connections of a single user due to failure of hardware components in the management system is LOW. A failure of hardware components in the management system can cause downtimes and instability of certain functions or parameters. Since the 5G management system often relies on interconnection and exchange of information certain hardware failures can cause unavailability of the whole management system. Though, the management system itself has in a short and medium time no direct influence on the functionality of the core and RAN, and therefore the quality of the communication. On the other hand, a longer time period can, through the accumulation of errors, cause degradation and finally failure of communication services. Overall, the probability is considered as LOW.	Low

5.3.7 Connection Failure

According to the description of the risk scenario, connection failures are seen in a similar way to the case in which urgent telephone calls cannot be made. The difference in this scenario is that an existing connection is interrupted instead of the call not being able to be made at all.

Typically, the management system is not directly involved, unlike the RAN, where the connection could be masked via jamming on the radio interface, or the 5G core, whose explicit task is to maintain stable and continuous network connectivity.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of connection failure due to misconfiguration of the 5G management system is classified as LOW. Connection failure is possible if there is a network outage. As described for regional and multiregional network outages the probability of this being caused through management systems is low. Another case for which connection failure is possible is that the user is not allowed to connect to the network or has used all its call contingent. This could happen if someone changes the properties of the user in the management system. Unintentional changes in exactly these moments seem near impossible. Although such intentional attacks are also possible, it is not a common attack goal (as described in “urgent telephone calls”) and therefore very unlikely to happen, especially due to misconfiguration of the management. Therefore, the probability is LOW.	Low
Insufficient Access Controls	Availability (AP2-A)	The risk probability of connection failure due to insufficient access control of the management system is classified as LOW. Although insufficient physical or remote access control can empower internal and external attackers to have access to 5G management surfaces, effectively altering network configurations (e.g. handover parameters or switching off network elements) [SUPATK, HTROJ] to affect user connections is considered highly unlikely [E]. Therefore, the probability is LOW.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of connection failure due to implementation flaws in the 5G management system is classified as LOW. Implementation flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems. While unavailability of the MGMT system over extended periods will lead to accumulations of unresolved errors or security incidents which may finally cause service failure, a misconfigured MGMT in itself will not lead to an immediate failure of the service functionalities. In addition, intentional manipulation of software by e.g. supply chain attack [SOLAR] is a possible attack path. But since such attacks have high complexity and resource requirements they are very difficult to carry out. The overall probability of the risk is therefore considered LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Availability (AP4-A)	The risk probability of connection failure due to specification weaknesses of the management system is classified as LOW. Without an official specification for 5G management, implementation errors in individual systems could limit the management's availability or increase the likelihood of attacks (e.g. denial of service). Limiting the availability of the system could have a negative impact on connections to the extent that necessary (re-)configurations of network parameters cannot be carried out (e.g. neighbour relationships for handovers), which could ultimately also affect active connections. However, this scenario has a greater impact on general connectivity problems and less on specific user connections. In addition, these problems arise in the event of longer-term outages, which are usually very unlikely with management systems. Therefore, the probability is LOW.	Low
Failure of Power Supply	Availability (AP5-A)	The risk probability of connection failure due to failure of power supply in the 5G management system is classified as LOW. Connection failures in 5G systems may occur due to an outage of the management system. However, such outages are highly unlikely as there normally is little to no dependency of 5G core system on the management system regarding running connections. In addition to that, as stated earlier, the management system of public MNOs normally runs on very resilient systems. Therefore, the probability LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of connection failure due to failures of hardware components of the management system or the unavailability of spare parts is classified as LOW. Although most identified attack paths in this category target the availability of the management system, the availability of active connections is not directly affected. Similar to attacks targeting the availability of the management system, necessary (re-)configurations of network parameters cannot be carried out in the event of long-term outages. However, redundancy concepts are typically used for critical systems. In addition, mobile networks are typically robust for some time during which no interventions can be applied [ENISAINC, E]. Therefore, the probability is LOW.	Low

5.3.8 Tapping or Altering of the Geographical Position

For the tapping or altering of the geographical position we consider the following: The management system is not involved in estimating users' positions and does only show the general positions of users over PCIs, and TACs. However, as already stated in the risk scenario of telecommunications secrecy, the management system has access to user's metadata, e.g., connections made by the UE to RUs. User identities such as the IMEI of the UE may be linked to the IMSI potentially leaking the identity and more detailed geographical position, e.g. through profiling.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The probability of an attacker being able to do tapping or altering of the geographical position of users due to misconfiguration in the management system is considered MEDIUM. Note that altering the geographical data on the management system does not affect the operation of the underlying network. This stems from the fact that management systems only display information about the UEs position but cannot be used to return altered positioning data to the UE. Another attack path that targets the altering and tapping of geographical position of users is related to displaying of manipulated log data. An example could be the exfiltration and manipulation through the lawful interception system [ATHEN], or the software that displays this information on management systems. Additionally, tapping can occur through wrongly configured API endpoints [MAIL]. Tapping and altering are possible through weak/wrongly configured APIs or internal software and applications [TELEAPI2] [TELEAPI]. Tapping is confidentiality related and of MEDIUM probability as it is similar to telecommunication secrecy, whereas altering is related to integrity and has a LOW probability.	Medium
Insufficient Access Controls	Confidentiality (AP2-C)	The probability of tapping or altering the geographical positioning of users enabled by insufficient access control is categorized as MEDIUM. Initial access to the necessary management systems could most likely be gained due to weak authentication abuse [SECSPEC] [ATNT24] [Kyiv], privilege escalation or simply insufficient access control and credential theft [BSI_Threat] [ENISA], for example an insider using an unlocked laptop of a colleague to update his own service settings. Another attack path could be injection attacks due to insecure APIs [SEEAPI]. Any attacker that has access to relevant security logs and GUI of management system can tap into substantial information about the geographical position of any device in the network, assuming other linking information about the targeted UE is in the hand of the attacker (e.g. IMSI, IMEI, names). The probability is also increased since no higher privilege escalation would be needed for tapping into this information. Tapping is confidentiality related and of MEDIUM probability, whereas altering is related to integrity and has a LOW probability.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The probability of tapping or altering the geographical position due to an implementation flaw in the management system is considered MEDIUM. This is because implementation flaws might open attack paths that allow an attacker to access location data. Examples that such implementation flaws may happen are [LOG4SH][HEARBL][ATHEN]. Especially the Athens affair is a good example on how implementation flaws can lead to a situation where confidential Information is leaked. Albeit the Athens attack did not use the management system, it would be possible to undertake similar attacks on the management system in order to gain access to geographical metadata. Tapping of geographical positions is a confidentiality related issue and of MEDIUM probability, whereas altering of such data is related to integrity and is much more complex and therefore has a LOW probability.	Medium
Specification Weakness	Confidentiality (AP4-C)	The probability of tapping or altering the geographical position due to a specification weakness in the management system is considered LOW. Since there is no specification for management systems every vendor creates their own system with their own architecture decisions. This in theory can lead to very unsecure systems e.g. where every user is able to tap or alter geographical positions. Since the geographical position is highly confidential data and insecure handling of such data is connected to high fines and legal problems for the MNOs they are expected to take good care of securing such data. Also, the implementation of lawful interception which is the main reason for such data being available in the management system in the first place, must be done carefully to prevent risks of data leakage. Even though errors are possible here, as seen in the Athens Affair [ATHEN], the probability of them occurring and being exploited to tap or alter geographical data is considered LOW.	Low
Failure of Power Supply	Confidentiality (AP5-C)	No relevant attack path could be identified. Since tapping or spoofing of geographical user position is done through the management system a failure of the power supply itself only damages its availability and in consequence the ability of tapping. Any offline tampering with the geographical data on the management system itself does not affect the network below since the management system is only displaying this information and there is no input of geographical location from the management system to the network.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	No relevant attack path could be identified. As described above, any failure that causes unavailability of the management systems does not enable the tapping or tampering of the geographical position of a user since the management system receives the information from underlying systems.	Low

5.4 Private 5G Network Operator Perspective

Private networks, while offering various benefits, present distinct challenges concerning security. These networks often employ non-carrier-grade hardware and software, which may be more prone to implementation flaws. Unlike public networks, private systems are exempt from the obligation to incorporate lawful interception capabilities, potentially diminishing their overall security framework. Additionally, because private networks integrate into a company's local IT infrastructure and avoid communication via the public internet, they reduce exposure to internet-based attacks. Nonetheless, they remain vulnerable to traditional IT attacks. The responsibility for operating these networks lies with the user company, necessitating a high level of expertise from IT staff who may lack telecommunications experience. This requirement increases the risk of human error. Conversely, the absence of billing and customer service functions in private networks means there is typically no need to store confidential personal data, thereby mitigating risks associated with data breaches.

5.4.1 Network Failure

In a private 5G network, a network failure is typically a local small-scale issue in opposite to the public network. The 5G management system is normally deployed and operated on standard server infrastructure within the company premise. Therefore, private companies, that are well experienced in operation of IT systems, may replicate this knowledge in regard to the 5G management system, in order to guarantee a reliable operation. Additionally, the isolated deployment of the 5G network makes it less prone to attacks from outside sources. Still, the network may be negatively influenced by internal sources.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of network failure due to misconfiguration of the management system from the perspective of a private network operator is considered LOW. In our analysis misconfiguration of the management system only relates to the configurations of the management system itself. So configurations of the underlying components like RAN and core network are not considered. Therefore, the worst impact due to misconfiguration of the management is that the management system is not working properly or is even completely unavailable. In this situation the management system can't be used to configure the network, which may lead to problems. Since the 5G network is running autonomously without updates to its configuration, once it has been initially deployed, the risk of a misconfiguration leading to network failure is considered LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	The risk probability of network failure due to insufficient access control in the management system from the perspective of a private network operator is considered MEDIUM. Insufficient access control in a 5G management system can allow attackers to have access to administrative information and controls. Exploiting weak authentication, insecure APIs, and physical access vulnerabilities, attackers can disrupt network operations, especially by misconfiguration of underlying sub-systems. A compromised MGMT system in a private network will not directly lead to network failure. However, for a private network operator, this could result in network failures due to unauthorized control and manipulation of network resources. Given the likelihood of such attacks in environments lacking carrier-grade security, the risk probability is MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The risk probability of network failure due to implementation flaws of the management system in a private network is LOW. As mentioned on the preface, private 5G networks are more prone to implementation flaws because of varying quality of software development. These flaws might render the management system completely unavailable, because of typical software development issues like e.g. null pointer exception, or negatively impact the performance of such systems. This may hinder the general usability of the management system itself and connected systems. Because of the separation of management and execution, an outage of the management system won't directly impact the availability of a network. Therefore, the probability is considered LOW.	Low
Specification Weakness	Availability (AP4-A)	The risk probability of the risk that weaknesses in the specifications will cause a network failure from the perspective of a private network operator is classified as LOW. The specifications for the management system are almost non-existent, so it is very unlikely that serious conflicts will be incorporated. Availability is an important objective for telecommunications networks, to which operators pay particular attention. Hence, they design their management system implementation to promote system availability.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The network failure risk probability due to failure of power supply of the management system is LOW. Failure of power supply may be caused by natural disasters, accidents or terrorist attacks. A failure of power supply might render the management system unavailable, preventing a private operator from applying new configurations. However, it might take a relatively long time before the inaccessibility of the MGMT system leads to network problems, because of the autonomy of 5G systems. What is to be considered here is, that for private networks the regional expansion of the computing facilities will be rather small. So an outage will most of the time not only affect the MGMT system but also the other components. Since it is considered that only the MGMT system is affected and the rest of the 5G system can continue operation, the probability for this outage of the MGMT happening and also affecting the other components through the absence of configuration ability is considered LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The risk probability of network failure due to failure of management hardware components / unavailability of spare from the perspective of a private network operator is LOW. The risk of failure of networks in a 5G management system can arise from various hardware-related issues. Failure of MGMT HW won't lead to immediate outage of the cellular system, but unavailability of the MGMT system over extended periods will lead to accumulation of unresolved errors or security incidents, that might not be recognized and will therefore increase the probability of failure. In the end, this could lead to a failure of the system. In this situation, the longer the system is unavailable, the higher the probability of failure will be. Additionally, in the case of a private network, redundancy and availability of spare parts might not be comparable to a carrier-grade system, but normally private network operators are sensitised to IT-Network operation and the procedures that go along. Hardware might also be shared amongst multiple components of the system, e.g. Core and MGMT run on the same hardware [E]. However, by incorporating this risk into the planning process, a local private network can operate with sufficient redundancy at hand reducing the probability of the risk to LOW. Normally, there is no special hardware needed for the MGMT system and the resource requirements are rather low. Therefore, unavailability of hardware means unavailability of a medium grade COTS server, which is very unlikely.	Low

5.4.2 Degraded Services / Connections

Degraded services or connections in a private network can be caused by an issue with the 5G management system of the private operator. Network configuration is administrated by local IT personnel who may not have access to an extensive testing infrastructure. On the one hand, this can lead to the deployment of insufficiently tested configurations, resulting in degraded network performance. On the other hand, even with carefully tested configurations and separated deployments, errors due to a malicious user or unintended configuration might degrade services and connections of the private 5G network. Another issue with private networks is that non-specialist personnel might be operating and maintaining the 5G network. This can lead to an increased probability of human errors affecting the stable and secure operation of private networks.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk probability of degraded services or connections due to misconfiguration in the management system within the private 5G network operator perspective is considered MEDIUM. The general service and prioritization of network traffic can be influenced over the management system on a larger scale through quality of service and user group settings. These can cause a downstream effect that change or influence core, RAN and finally user settings which result in slowing down of services. This is because configurations of user or service quality class details and parameters are likely done by users or administrators, that are less experienced in the management of private 5G network systems. Furthermore, since the impact is only companywide, there will often be less mechanisms preventing harmful configurations compared to public networks. Other misconfiguration can cause delays or unavailability during the update of relevant parameters, for example endpoints misconfigurations or more likely energy saving settings. In addition, as always, misconfigurations of the security settings can be exploited by attackers.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	The probability of degraded services or connections of the network due to insufficient access control of management systems within the private 5G network operator perspective is considered MEDIUM. Initial access to the necessary management systems could most likely be gained due to weak authentication abuse [SECSPEC] [ATNT24] [Kyiv], privilege escalation or simply insufficient access control and credential theft [ENISATHREAT], for example an insider using an open laptop of a colleague to update his own service settings. Injection attacks due to insecure APIs [SEEAPI] could also be utilized for unauthorized access to the management system. Once initial access is gained different misconfigurations, as seen above, can be executed. With multiple attack paths available, the probability is evaluated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The probability of degraded services or connections in the network due to an implementation flaw in the management system within the private 5G network operator perspective is considered MEDIUM. A disruption of an attacker that affects the management system's availability has in the short and medium time span no effect on the quality of service of the underlying communication of the mobile network. As a consequence, only implementation flaws that affect the integrity of overlying parameters that steer the quality of the communication itself are relevant. Next to zero-day exploits that allow the tampering of data over secure protocols [LOG4SH] [HEARBL], especially outdated external software [PETYA] that is used for management software systems may enable attackers to tamper with subscriber data for quality-of-service assurance. Other than that, supply chain attacks [SOLAR] can be launched to create implementation errors that change subscriber and quality of service data in a similar sense. Recent occurrences of such attacks evaluate the probability to MEDIUM.	Medium
Specification Weakness	Availability (AP4-A)	The probability of degraded services or connections in the network due to specification weakness in the management system within the private 5G network operator perspective is LOW. The lack of official specifications for 5G management systems can lead to inconsistent implementation of security measures and architectural best practices by different vendors. This inconsistency may result in systems exhibiting varied behaviors, potentially compromising all security goals, including those related to service and connection handling. While weaknesses might be extrapolated through the missing of specifications that can cause DoS or insufficient access control it does not directly influence the quality of service of the communication of the underlying network.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The probability of degraded services or connections in the network due to a power outage that takes down parts or the whole management system within the private 5G network operator perspective is LOW. Failure of power supply may be caused by natural disasters, accidents or terrorist attacks. It is considered that a failure of the power supply of the 5G management system will only cause unavailability of this system. This assumption stems from the fact that for private networks the regional expansion of the computing facilities will be rather small. An outage will most of the time not only affect the MGMT system but rather nearly all system components. But since the case where the power supplies of the other components fail is considered separately in the corresponding analyses, it is not evaluated here. Changes or manipulation of the underlying data due a power outage is highly unlikely. Also, the management system itself has no direct influence on the functionality of the core and RAN in a short and medium time and, therefore, on the quality of the communication. In a longer period, the accumulation of errors can cause degradation and finally failure of communication services. But since the probability for a power outage lasting that long is very low. Although there might not be any extra redundancy in private networks, that can be used in case of failure of the main power supply, the overall risk is still considered as LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of degraded services or connections in the network due to hardware failures in the management system within the private 5G network operator perspective is LOW. A failure in the hardware of the management system can cause the system to become unavailable. Though, the management system itself has in a short and medium time no direct influence on the functionality of the core and RAN, and therefore the quality of the communication. On the other hand, a longer time period can, through the accumulation of errors, cause degradation and finally failure of communication services. But since it is very unlikely for a hardware failure to cause long lasting outage of the management system, especially in the case of a private network, where no special hardware will be used and every part can easily be replaced, the overall risk is considered as LOW.	Low

5.4.3 Information Disclosure

In the description of this risk scenario is stated that information disclosure in this case is only related to the transport layer and the associated encryption. Metadata or other information is not considered here. As no communication is routed through the management system, information disclosure is not directly possible due to an attack on the management system. It is possible though, that the attack on the management can lead to a manipulation of configuration of the underlying components, which can lead to information disclosure in the described sense. However, this is not considered in the management system analysis as stated earlier.

5.4.4 User Identities (Authenticity/Integrity)

This risk scenario is about an attacker being able to spoof identities to remove authentication and integrity of messages sent over the network to users. A user identity in current 5G systems is mostly authenticated during a procedure called Authentication and Key Agreement (AKA). While there might be alternative authentication protocols, currently 5G authentication is based on the 5G AKA and EAP-AKA', which use mutual authentication through two shared permanent keys and UE identity. The SUPI is hereby the identifier of the UE, just as the IMSI in former generations, that is used to initiate the authentication process with the 5G home network. This SUPI can simply be viewed by anyone that has access to the MGMT system. The permanent keys, shared by the UE and the home network, on the other hand cannot simply be accessed in the MGMT system. These keys are also used to derive keys for integrity protection.

To masquerade or spoof a user identity in a 5G network the two permanent keys would need to be eavesdropped. They rest in the physical SIM of the UE and the 5G Core function UDM (or HSS for 4G Core systems). The management system generally has no direct access to either of these. However, depending on the management system of the core, the provisioning of the keys is done through the management system. If this process is eavesdropped this can potentially lead to a spoofing of user identities. Furthermore, since the exact functions of the MGMT systems and how they are implemented are not publicly available, it is uncertain if there exists a MGMT that can read out the keys after provisioning. Generally, this is not the case though, and to be clear it is not possible to read key information after provisioning and it will therefore not be considered here [E].

This only leaves three attack paths, which are highly sophisticated and/or need complete access to the MGMT system. The first attack path simply includes a DoS attack on any UE by tampering with the users stored in the core. The second includes malicious provisioning, by adding a new user to the network which enables an attacker to add malicious devices to the network. These two attack paths are not relevant for the evaluation of this risk scenario because it is not about DoS or infiltration but precisely about spoofing a user identity. Lastly, the third attack path can be considered relevant to the observed risk scenario and is the one that is evaluated here. In this case the permanent shared keys are eavesdropped during the provisioning and then used in conjunction with a new SIM card and a malicious UE. It is to note that this case is very difficult and sophisticated.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The risk probability of misconfiguration in 5G management systems impacting user identities (authenticity/integrity) is classified as LOW. Accessing the underlying 5G management systems like the UDM within the core domain enables tampering with permanent user identifiers directly. These misconfigurations can compromise the authenticity and integrity of user identities. While this can be considered part of the core risk analysis it will be mentioned briefly in the following. In general, there are the three attack paths related to tampering with the UDM in the core that were mentioned in the preface, that may be abused during provisioning process in the management system. The first is causing unavailability to the UE by changing or deleting its public key in the network. The second is performed by adding or reconfiguring of an existent public key. In a later step, this would be used to connect to the public network and masquerade as a new device that can access all the information and services the legitimate UE can. Furthermore, the third attack path, observing the provisioning process, cannot be executed through misconfigurations. Since the two relevant attack paths for misconfigurations are not directly covered by the risk scenario of stealing user identities, we evaluate the risk probability as LOW. Still, it is worth to mention other related risks, even if they do not directly affect this risk scenario.	Low
Insufficient Access Controls	Integrity (AP2-I)	The probability of authenticity/integrity loss of user identities due to insufficient access control within the private 5G network operator perspective is LOW. Insufficient access control in a 5G MGMT system can pose a significant risk for authenticity/integrity of user identities. An attacker exploiting weak authentication, stealing credentials, or escalating privileges can access sensitive data [SECSPEC]. Once access is gained to the management system an attack would follow the attack path described in the preface above. Overall, we evaluate the probability risk of a malicious attacker accessing management system, eavesdropping on provisioning or tampering with authentication related data as LOW.	Low
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	The risk probability of implementation flaws in the 5G management system affecting user identities is classified as LOW. User Identities are only accessible by the management system during the small timeframe of the provisioning process. Because most flaws are unintentional and lead to system errors or unpredictable data leakage, this kind of identity leakage is highly unlikely. From a current perspective private networks often employ non-carrier-grade hardware and software, which may be more prone to implementation flaws. Overall, the probability of this attack path is evaluated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Integrity (AP4-I)	The probability of the authenticity/integrity of user identities being compromised due to specification weaknesses from the perspective of private 5G operators is LOW. In the 5G management system, specification weaknesses, such as missing or insufficient protocol definitions between management components (such as TLS) can significantly impact user integrity [SECSPEC]. For a private 5G network operator, these vulnerabilities can lead to unauthorized access to the 5G network. The probability of this risk is low due to the typically robust security measures, state-of-the-art protocols and architectures of communication protocols. Furthermore, focusing on on-premise networks, private network operators typically will have higher isolation from external attacks that leads to less possible exploits of potential specification weaknesses.	Low
Failure of Power Supply	Integrity (AP5-I)	No relevant attack path could be identified.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	No relevant attack path could be identified.	Low

6 Risk Assessment Results

This chapter presents the final results of the detailed risk analysis.

Note: Risk scenarios with no given probability level were not rated, see beginning of Chapter 5 and the corresponding risk scenario.

6.1 Probability Assessment (isolated)

The following table summarizes the results of this detailed analysis of the **management system**. According to the maximum principle in risk analysis only the highest probability is listed.

Note: As described in Chapter 5, attacks which are based on changing or manipulating the configuration of the 5G components itself (with the help of the compromised management system) were not assessed since this kind of misconfigurations are discussed in the detailed risk analyses of the subsystems Core, RAN, Transport, IMS/Interconnect, Infrastructure, and MEC.

Perspective	Section	Risk Scenario	Probability Level
Public MNO	5.2.1	Emergency Call (TKG § 164)	Low
	5.2.2	Cell Broadcast (TKG § 164a)	Low
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	Medium
	5.2.4	Emergency Preparedness (TKG § 184)	Low
	5.2.5	Regional Network Failure	Low
	5.2.6	Multi-Regional Network Failure	Low
	5.2.7	Failure of Mobile Service	Medium
	5.2.8	Fraud	Medium
	5.2.9	Subscriber-Data Leak	Medium
User	5.3.1	Urgent Telephone Calls Not Possible	Low
	5.3.2	Authenticity of the Receiver of a Call	Low
	5.3.3	Integrity of Transmitted Data	
	5.3.4	User's Identity	Low
	5.3.5	Eavesdropping of a Communication	Low
	5.3.6	Slowing Down/Restricting the Connection	Medium
	5.3.7	Connection Failure	Low
	5.3.8	Tapping or Altering of the Geographical Position	Medium
Private 5G Network Operator	5.4.1	Network Failure	Medium
	5.4.2	Degraded Services/Connections	Medium
	5.4.3	Information Disclosure	
	5.4.4	User Identities (Authenticity/ Integrity)	Low

6.2 Probability Assessment (merged)

Since management systems are essential for configuration and operation of all subsystems in a mobile network, the probability assessment regarding intentional or unintentional misconfiguration of the subsystems Core, RAN, Transport, IMS/Interconnect, Infrastructure, and MEC are merged with the results of this risk analysis. The following table summarizes the probability assessment in terms of misconfigurations and shows the **merged probability level for the management subsystem**. According to the maximum principle in risk analysis only the highest probability is listed. The merged probability level of the management system is then used to build the risk score in section 6.3.

Perspective	Risk Scenario	Probability Level of Misconfiguration							Management (merged)
		Core	RAN	Transport	IMS and Interconnect	Infrastructure	MEC	Management (isolated)	
Public MNO	Emergency Call (TKG § 164)	Medium		Medium	Medium			Low	Medium
	Cell Broadcast (TKG § 164a)	Low	Low	Low				Low	Low
	Telecommunications Secrecy (TTDSG § 3)	Medium	Medium	Low	High	High	Medium	Medium	High
	Emergency Preparedness (TKG § 184)	Medium	Low	Medium	High			Low	High
	Regional Network Failure	Low	Medium	Medium				Low	Medium
	Multi-Regional Network Failure	Low		Medium		Medium		Low	Medium
	Failure of Mobile Service	Medium		Medium	Medium		High	Medium	High
	Fraud	Medium			High			Medium	High
	Subscriber Data Leak	Medium	Low	Low	High	High		Medium	High
User	Urgent Telephone Calls Not Possible	Medium	Medium	Medium	Medium			Low	Medium

Perspective	Risk Scenario	Probability Level of Misconfiguration						Management (isolated)	Management (merged)
		Core	RAN	Transport	IMS and Interconnect	Infrastructure	MEC		
	Authenticity of the Receiver of a Call	Medium			Low			Low	Medium
	Integrity of Transmitted Data	Medium	Medium	Medium	Medium	Medium	Low		Medium
	User's Identity	Medium	Low		Medium	High	Medium	Low	High
	Eavesdropping of a Communication	Medium	Low	Low	Medium	Medium	Medium	Low	Medium
	Slowing Down/Restricting the Connection	Medium	Medium	Medium	High		High	Medium	High
	Connection Failure	Medium	Medium	Medium	High			Low	High
	Tapping or Altering of the Geographical Position	Medium	Low		Medium		Low	Medium	Medium
Private 5G Network Operator	Network Failure	Low	Medium	Low		Medium	Medium	Medium	Medium
	Degraded Services / Connections	Low	High	Medium		High	Medium	Medium	High
	Information Disclosure	Medium	Medium	Medium			Medium		Medium
	User Identities (Authenticity/Integrity)	Medium	Medium				Low	Low	Medium

6.3 Risk Assessment

The following table summarizes the overall risk assessment of the **management system**. The risk scenarios and their respective impact level are described in the framework document.

Perspective	Risk Scenario	Impact	Probability	Risk Score
Public MNO	Emergency Call (TKG § 164)	High	Medium	High
	Cell Broadcast (TKG § 164a)	High	Low	Medium
	Telecommunications Secrecy (TTDSG § 3)	Medium	High	High
	Emergency Preparedness (TKG § 184)	High	High	High
	Regional Network Failure	Medium	Medium	Medium
	Multi-Regional Network Failure	High	Medium	High
	Failure of Mobile Service	Medium	High	High
	Fraud	Low	High	Medium
	Subscriber-Data Leak	Medium	High	High
User	Urgent Telephone Calls Not Possible	Low	Medium	Low
	Authenticity of the Receiver of a Call	Medium	Medium	Medium
	Integrity of Transmitted Data	High	Medium	High
	User's Identity	High	High	High
	Eavesdropping of a Communication	Medium	Medium	Medium
	Slowing Down/Restricting the Connection	Low	High	Medium
	Connection Failure	Low	High	Medium
	Tapping or Altering of the Geographical Position	High	Medium	High
Private 5G Network Operator	Network Failure	Medium	Medium	Medium
	Degraded Services/Connections	Medium	High	High
	Information Disclosure	Medium	Medium	Medium
	User Identities (Authenticity/ Integrity)	Medium	Medium	Medium

Bibliography

[E]	Expert Knowledge
[ATHEN]	Athens Affair https://spectrum.ieee.org/the-athens-affair
[BNETZA]	Legislative framework for emergency calls, 21.01.2014 https://www.bundesnetzagentur.de/EN/Areas/Telecommunications/ServiceProviderObligation/EmergencyCalls/LegislativeFramework/LegislativeFramework_node.html
[ENISATHREAT]	“ENISA threat landscape for 5G Networks.” Accessed: May 14, 2024. [Online]. Available: https://www.enisa.europa.eu/publications/enisa-threat-landscape-for-5g-networks
[ENISAINC]	ENISA, “Telecom Security Incidents 2022”. Mar. 2024. https://www.enisa.europa.eu/publications/telecom-security-incidents-2022
[ESEC]	D. Sabella, “MEC security; Status of standards support and future evolutions,” Sep. 2022.
[SECENG]	Ross, Anderson, Security Engineering, 2nd ed. Wiley, 2008.
[CYSEC]	Ahmed et al. (2021). Cyber Security in IoT-Based Cloud Computing: A Comprehensive Survey. Electronics. 16. 10.3390/electronics11010016.
[SUPATK]	D. Fiser, “Supply Chain Attacks in the Age of Cloud Computing: Risks, Mitigations, and the Importance of Securing Back Ends,” 2020.
[NSA]	Nick Sullivan, “How the NSA (may have) put a backdoor in RSA’s cryptography: A technical primer”, 2014, https://arstechnica.com/information-technology/2014/01/how-the-nsa-may-have-put-a-backdoor-in-rsas-cryptography-a-technical-primer/
[SUEZ]	David Welch, “The Suez Canal And Our Fragile Global Supply Chain: How Redundancy In Industrial Development Can Ease The Strain,” Forbes, May 2021. https://www.forbes.com/sites/forbesrealestatecouncil/2021/05/04/the-suez-canal-and-our-fragile-global-supply-chain-how-redundancy-in-industrial-development-can-ease-the-strain/
[STUXNET]	Thomas M. Chen et al., “Lessons from Stuxnet”, IEEE Computer, 2011. https://doi.org/10.1109/MC.2011.115
[DOSCONT]	Raja Ettiane et al. “Toward securing the control plane of 5G mobile networks against DoS threats: Attack scenarios and promising solutions”. 2021. https://doi.org/10.1016/j.jisa.2021.102943
[SECSPEC]	Roger Piqueras Jover et al. “Security and Protocol Exploit Analysis of the 5G Specifications”. 2019. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8641117
[HUA]	Philipp Eckstein “Bund verbietet Huawei-Komponenten im 5G-Netz” https://www.tagesschau.de/inland/huawei-5g-verbieten-100.html

[MICRO]	Eva Georgieva „2.4 TB Data Leak Caused By Microsoft’s Misconfiguration“, 2022 https://purplesec.us/security-insights/microsoft-data-leak/#:~:text=Misconfiguration%20of%20an%20endpoint%20caused%20a%20leakage%20of%202.4%20TB%20of%20data%20of%20Microsoft%E2%80%99s%20customers
[MAIL]	VeraSafe, „Accidental Data Breach? Misdirected Emails Can Land You in Hot Water“, 2024 https://verasafe.com/blog/misdirected-emails-prevention-legally-required/
[SSLTLS]	Jack O'Sullivan, “Understanding Common SSL/TLS Misconfigurations”, 2021 https://secarma.com/understanding-common-ssl-tls-misconfigurations-2/#:~:text=SSLv2%20%28Deprecated%29
[NISSAN]	Catalin Cimpanu, “Nissan source code leaked online after Git repo misconfiguration”, 2021 https://www.zdnet.com/article/nissan-source-code-leaked-online-after-git-repo-misconfiguration/
[TS38331]	NR; Radio Resource Control (RRC); Protocol specification, v18.1.0 https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3197
[TS24501]	Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3, v18.6.0 https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3370
[TS33501]	Security architecture and procedures for 5G System, v18.6.0 https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3169
[GOOCL]	“Incident affecting Google Cloud Networking, Cloud Run, Google App Engine, Google Cloud Functions, Apigee, Cloud Load Balancing” https://status.cloud.google.com/incidents/6PM5mNd43NbMqjCZ5REh#:~:text=On%20Tuesday%2C%2016%20November%202021%20at%2009%3A35%20PT%2C,response%20to%20HTTP%2FS%20requests.
[OWASP]	OWASP Top Ten https://owasp.org/www-project-top-ten/
[HEARBL]	The Heartbleed Bug https://heartbleed.com/
[LOG4SH]	Java-Bibliothek Log4j – eine Bilanz https://www.bsi.bund.de/DE/Themen/Verbraucherinnen-und-Verbraucher/Cyber-Sicherheitslage/Schwachstelle-log4Shell-Java-Bibliothek/log4j_node.html
[WANCRY]	WannaCry https://www.bsi.bund.de/DE/Themen/Verbraucherinnen-und-Verbraucher/Cyber-Sicherheitslage/Methoden-der-Cyber-Kriminalitaet/Botnetze/Steckbriefe-aktueller-Botnetze/Steckbriefe/Wannacry/Wannacry.html

[PETYA]	Was sind Petya und NotPetya, Lars Becker, IT Verlag GmbH, https://www.it-daily.net/it-sicherheit/cybercrime/petya-notpetya
[EQUIFAX]	Equifax data breach FAQ: What happened, who was affected, what was the impact? https://www.csoonline.com/article/567833/equifax-data-breach-faq-what-happened-who-was-affected-what-was-the-impact.html
[DIRCOW]	Dirty COW https://en.wikipedia.org/wiki/Dirty_COW
[SOLAR]	Ein Hackerangriff, der um die Welt geht https://www.spektrum.de/news/solarwinds-ein-hackerangriff-der-um-die-welt-geht/1819187
[ARIANE]	Ariane flight V88 https://en.wikipedia.org/wiki/Ariane_flight_V88
[AMA14]	Amazon, Summary of the Amazon EC2, Amazon EBS, and Amazon RDS Service Event in the EU West Region, 2014, https://aws.amazon.com/de/message/2329B7/
[AMA16]	Amazon, Summary of the AWS Service Event in the Sydney Region, 2016, https://aws.amazon.com/de/message/4372T8/
[GOO15]	Google, Google Compute Engine Persistent Disk issue in europe-west1-b, 2015, https://status.cloud.google.com/incident/compute/15056#5719570367119360
[ATNT24]	Massive AT&T hack compromises phone, text records of "nearly all" customers, 2024. https://about.att.com/story/2024/addressing-illegal-download.html
[GTPDOOR]	GTPDOOR - A novel backdoor tailored for covert access over the roaming exchange, https://doubleagent.net/gtpdoor-a-novel-backdoor-tailored-for-covert-access-over-the-roaming-exchange/
[SOKFRAUD]	M. Sahin, A. Francillon, P. Gupta, and M. Ahamad, "SoK: Fraud in Telephony Networks," in 2017 IEEE European Symposium on Security and Privacy (EuroS&P), Apr. 2017, pp. 235–250. doi: 10.1109/EuroSP.2017.40.
[SECTIGO]	Nick France, The risks of expired SSL certificates for enterprise organizations, 2024, https://www.sectigo.com/resource-library/industry-impact-expired-ssl-certificate-outages
[TELEAPI]	Marc Stöckel, API ermöglicht Tracking von Festnetzanschlüssen, 2024, https://www.golem.de/news/datenpanne-bei-der-telekom-api-ermoeglicht-tracking-von-festnetzanschlussen-2407-187066.html
[TELEAPI2]	Sergiu Gatlan, T-Mobile hacked to steal data of 37 million accounts in API data breach https://www.bleepingcomputer.com/news/security/t-mobile-hacked-to-steal-data-of-37-million-accounts-in-api-data-breach/
[TAGSTOE]	Tagesschau, Weltweit Störungen bei Computersystemen, 2024, https://www.tagesschau.de/ausland/it-panne-computer-global-100.html

[KYIV]	Cyberattack on Kyiv Star https://www.reuters.com/world/europe/russian-hackers-were-inside-ukraine-telecoms-giant-months-cyber-spy-chief-2024-01-04/
[SOLAR]	SolarWinds hack explained, Saheed Oladimeji, Sean Michael Kerner, 03.2023 https://www.techtarget.com/whatis/feature/SolarWinds-hack-explained-Everything-you-need-to-know
[CALLFOR]	Telecom operators are warning people about call-forwarding scams https://economictimes.indiatimes.com/news/new-updates/telecom-operators-are-warning-people-about-call-forwarding-scams-heres-how-to-protect-yourself/articleshow/104757879.cms?from=mdr
[EDSNOW]	NSA manipuliert per Post versandte US-Netzwerktechnik. Heise Online. http://www.heise.de/newsticker/meldung/NSA-manipuliert-per-Post-versandte-US-Netzwerktechnik-2187858.html
[HTROJ]	Yoshimura, M. & Ogita, Amy & Hosokawa, Toshinori. (2013). A smart Trojan circuit and smart attack method in AES encryption circuits. 278-283. 10.1109/DFT.2013.6653619.
[REDHAT]	The state of Kubernetes security report: 2024 edition https://www.redhat.com/en/engage/state-kubernetes-security-report-2024
[STUX]	Baezner, Marie & Robin, Patrice. (2018). Stuxnet, https://www.researchgate.net/publication/323199431_Stuxnet
[SEEAPI]	An open API at Deutsche Telekom is leaking user data https://borncity.com/win/2024/07/15/an-open-api-at-deutsche-telekom-is-leaking-user-data/
[CATCH]	IMSI-Catcher: Überwachung auch in 5G-Netzen möglich https://www.heise.de/news/IMSI-Catcher-Ueberwachung-auch-in-5G-Netzen-moeglich-9190322.html
[CROWDS]	2024 CrowdStrike incident https://en.wikipedia.org/wiki/2024_CrowdStrike_incident
[SYSLOG]	Syslog Flood Prevention https://www.manageengine.com/network-configuration-manager/help/syslog-flood-prevention.html
[NSAMEMORY]	National Security Agency Cybersecurity Information Sheet https://media.defense.gov/2022/Nov/10/2003112742/-1/-1/0/CSI_SOFTWARE_MEMORY_SAFETY.PDF
[XZ]	Kritische Backdoor in XZ für Linux https://www.bsi.bund.de/SharedDocs/Cybersicherheitswarnungen/DE/2024/2024-223608-1032.pdf?__blob=publicationFile&v=3
[OVH24]	Cloud-Dienstleister OVH: Feuer zerstört Rechenzentrum, ein weiteres beschädigt https://www.heise.de/news/OVH-Feuer-zerstoert-Rechenzentrum-in-Strassburg-ein-weiteres-beschaedigt-5076320.html

[BSIRZHV]	Sicherheit von Rechenzentren / Hochverfügbarkeit https://www.bsi.bund.de/DE/Themen/Unternehmen-und-Organisationen/Informationen-und-Empfehlungen/Empfehlungen-nach-Angriffszielen/Hochverfuegbarkeit/hochverfuegbarkeit_node.html
[GDPREU]	Data protection and working remotely https://gdpr.eu/working-remotely-data-security/
[INTERNAL]	Lazarus uses insiders, also see KnowBe4, Harmony Protocol and DeFiance Capital incidents https://hacken.io/discover/lazarus-group/